NPTEL MCQ Revision Notes

Course: Sustainable Transportation Systems | **Instructor shown on slides:** Prof. Bhola Ram Gurjar, Dept. of Civil Engineering | Lectures 1–10

Source: only the 10 uploaded PDFs (lec 01.pdf … lec 10.pdf). No outside knowledge added. Tags like `L4·s12` = Lecture 4, slide 12. Text in *[brackets]* is a note about the source. Yellow boxes = extraction issues.

Uploaded PPTs and contents

File	Lecture title (as on slide 1)	Slides
lec 01.pdf	L1: Introduction to Transportation Systems-I (evolution/timeline of modes)	50
lec 02.pdf	L2: Introduction to Transportation Systems-II (modes, infrastructure, efficiency, components, multimodal)	38
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Page guide: Part 2 = lecture-wise notes (L1→L10, each starts on a new page) · Part 2A = content read from image-only slides · Part 3 = rapid-revision sheet · Part 4 = 30 practice MCQs + answer key · Part 5 = coverage audit.

> **Note on image slides:** Many slides are pictures/charts with no text layer. I rendered and read them; their content is in **Part 2A** (after Lecture 10 notes). Values marked "≈" are approximate readings of unlabeled bars/lines. Not recoverable: Lecture 10 slides 11, 12, 14, 15, 16 (factor lists missing from the PDF). Slide numbers in each tag refer to the PDF page = slide.

Lecture 1 — Introduction to Transportation Systems-I (50 slides)

Topic: chronological evolution of transport modes. Most slides are photos with short captions; every dated fact is kept below.

Timeline table (in slide order)

Slide	Year / period	Fact (as on slide)
3	—	First form of transport on land = **walking**.
4	—	Donkeys, mules, horses used as transport for persons and goods.
5	**8000 BCE**	Primitive log rafts; tree trunk floating downstream may be the first stimulus to man's inventiveness. Oldest boat found = **Pesse Dugout Canoe** (8000 BCE).
6	**6000 BCE–3000 BCE** (Egyptian era)	Egyptians built some of the earliest boats recorded. First ones made of **papyrus reeds**, propelled by **rowing**; mainly used for **trade**. By **2500 BCE**, wooden boats that could cross oceans were built.
7	Viking era	Two classes: **war ships** ("**langskip**") and **merchant ships** ("**knorr**"). Fast, narrow war ships could survive ocean crossings and navigate very shallow water (draft as little as **50 cm / 20 in**). "One of the greatest technical and artistic achievement."
8	**3500 BC**	Wheel of Mesopotamia. Earliest wheels believed used for **pottery**.
9–11	—	Wheeled carts pulled for goods; animals trained to pull carts; horse-driven wagons, coaches, carriages = superior modes.
12	1800s	Zebra carts used by humans (humans tried to domesticate zebras).
13	**1490**	Leonardo da Vinci drew plans for a flying machine.
14	**1769**	**First true automobile**: Nicolas-Joseph **Cugnot**, three-wheeled steam-driven vehicle; heavy steam chamber in front → tendency to tip over when not hauling cannons (its design purpose). Steam buses running in Paris about **1800**.
15	**Dec 1, 1783**	Ballooning: French aeronauts **Jacques-Alexandre-Cesar Charles and Marie-Noel Robert** made first manned ascent in a **gas balloon**.
16	18th century	**Age of Canals**: first powerful motor of the industrial age; facilitated flows of goods, exchanges between regions, specialization of labor, access to distant energy/raw material. Local fuelwood shortages resolved by substituting **coal** (higher energy density), made possible by canals. Manayunk Canal section (Philadelphia) started in 1800s, operational up to the 20th century.
17	**1817**	**Draisienne** (Dandyhorse), **Karl Drais**: first **two-wheeled rider-propelled** machine, made of wood; rider paddled feet against the ground.
18	**1819**; 1825	Steam as fuel. **SS Savannah** = hybrid ship (steam engine + paddles using wind power); **first steamship to cross the Atlantic**. Steam-hauled public railways began with the **Stockton and Darlington Railway in 1825**. Reduced time to transport goods; allowed increased specialization.
19	**1825**	**Cheshunt Railway** = world's first **monorail**, drawn by a **horse**; overhead track, carriages suspended; "first passenger drawn Monorails in the world".
20	**1826**	**Omnibus** = first land-based innovation in public transportation system; horse-drawn passenger wagons.
21	**1830s–1930s**	**Great Railway Era**: first railways built in 1830s; extended range, speed, productivity achieved earlier by canals; new era of **coal, steam, steel, telegraph**; lasted till 1930s.
22	**1853**	George Cayley (English) designed glider; recorded as **first successful manned flight**.
23	**1860s**	**Velocipede**: bicycle version reinvented by the **Michaux family of Paris**; iron and wood, no springs → nickname "**boneshaker**"; driven by pedalling cranks on the **front axle**.
24	**1876**	First **steam-driven monorail**, demonstrated at US Centennial Exposition (Philadelphia); double-decker, two main wheels, rear one driven by rotary steam engine.
25	Title: **1876**; text: exhibited **1879** Berlin Commercial Exposition	World's first **electric tram** (Grob-Lichterfelde Tramway): driver seat + **3 little carriages, each 3 people**; ran on **direct current (approx. 149 volts)**. *[Slide title says 1876, body says 1879 — kept as written.]*
26	**1883**	**Penny-farthing** (James Starley): most used between the French mass-produced chain-driven **Boneshaker (1860s)** and the English **"safety bicycle" (1880s)**.

27	**Late 1885** / 1880s	**Rover Safety bicycle**, **John Kemp Starley**: forerunner of all modern bicycles; had **chain drive** → both wheels could be the **same size**.
28	1880s	Coal was world's primary heating and transportation fuel **late 1800s–1950s**. In **1880** coal powered a steam engine attached to the world's first electric generator. By late 1800s a new fuel: **petroleum**.
29	**1885**	**Daimler Reitwagen** = first true motorcycle (Gottlieb **Daimler** & Wilhelm **Maybach**, German); small petroleum/gasoline combustion engine on a wooden bicycle frame.
30	**1885**	First gasoline car. **Benz** ran first car 1885, **Daimler 1886**, **Hammel 1888**. First stationary gasoline engine by Carl Benz = **one-cylinder two-stroke**. "Benz Patent Motor Car" 1886.
31	**1896**	**Pilcher Hawk** (Percy Sinclair Pilcher): monoplane glider with birdlike wings.
32	1900s	Gasoline bus: nominal seating **20**, **four-cylinder** gasoline engine, **40 hp**, street speeds up to **32 km (20 mi) per hour**; built by **Mack Trucks**, sightseeing, Brooklyn's Prospect Park, 1900.
33	Early 1900s	**Airships/blimps**: inflated by **hydrogen** gas; engines with propellers + flaps to control direction and speed. (Zeppelin over Kiel, WWI.)
34	**1902** (Oct 24)	Wright brothers' **first fully controllable glider**, North Carolina.
35	**1905** (Oct 4)	**Wright Flyer** = first practical flying machine (Orville Wright; Huffman Prairie, near Dayton, Ohio).
36	**Apr 10, 1912**	**RMS Titanic** maiden voyage Southampton → New York City; world's largest passenger ship then; thought "unsinkable".
37	**1908**	**Ford Model T**: one of the first mass-production vehicles; turned automobile from luxury into necessity (cheap, versatile, easy to maintain).
38	**1919**	**Vickers Vimy**: John Alcock and Arthur Brown, **first nonstop transatlantic flight**.
39	**1935**	**Douglas DC-3** passenger aircraft first flew; dominated airline business until end of WWII.
40	**1959** (1959–2000)	**Hovercraft**: British-built/operated **air-cushion vehicles (ACVs)**; for **40 years** ferried passengers + automobiles across the English Channel (southern England ↔ northern France). Mountbatten class.
41	**1966**	**Bell Helicopter 206-B (Jet Ranger)**: turboshaft; carries **four** passengers; often used for deployment of firefighters.
42	**1969**	**Concorde**: first supersonic passenger-carrying commercial airplane (SST), built jointly by Great Britain and France. First flew **1969**, commercial service **1976**, operated till **2003**.
43	**June 5, 1969**	**Tupolev Tu-144** = first passenger aircraft to **break the sound barrier**.
44	1988	**Airbus A320**: short- to medium-range jetliner; first flew **1987**, commercial service **1988**; typically **150** passengers; derivatives **A318, A319, A321**.
45	**1999**	**Bombardier Global Express**: long-range business jet, twin engines; range > **11,100 km (6,900 mi)**; can approach speed of sound.
46	**2003**	**Maglev** (Transrapid 06, Germany trial): rides on air cushion created by electromagnetic reaction between on-board device and one embedded in guideway; propulsion and braking by varying **frequency and voltage of a linear motor** in guideway reacting with magnets on vehicle.
47	—	Timeline figure "Evolution of Transport Technology since the 18th Century" (Rodrigue 2020) — **image only**.
48	Conclusion	Transport systems changed drastically after the **Industrial revolution**. Coal and petroleum as major fuels had great impact on environment and natural resources. Need to reduce impact of emissions on environment.

Important Distinctions to Remember

- **Draisienne 1817** (first two-wheeler, wood, foot-paddling) ≠ **Velocipede 1860s** (Michaux, pedalling cranks on front axle, "boneshaker") ≠ **Penny-farthing 1883** ≠ **Rover Safety** (chain drive, wheels same size, forerunner of modern bikes).
- **Cheshunt 1825** = first monorail (horse-drawn) vs **1876** = first steam-driven monorail.
- **Cugnot 1769** = first true automobile (steam) vs **Benz 1885** = first gasoline car vs **Reitwagen 1885** = first true motorcycle.

- **Cayley 1853** = first successful manned flight (glider) vs **Wright glider 1902** (first fully controllable) vs **Wright Flyer 1905** (first practical) vs **Vickers Vimy 1919** (first nonstop transatlantic).
- **Concorde** (first supersonic passenger commercial airplane, UK+France) vs **Tu-144** (first passenger aircraft to break sound barrier, June 5 1969).
- Viking: war ship = **langskip**; merchant ship = **knorr**. Airships use **hydrogen**.

Image-only slides of this lecture were rendered and read; their content is in **Part 2A (end of Part 2) under Lecture 1**.

Lecture 2 — Introduction to Transportation Systems-II (38 slides)

Definitions L2·s3

- **Transportation system:** "the combination of elements and their interactions, which produce the **demand for travel** within a given area and the **supply of transportation services** to satisfy this demand." (Ennio Cascetta, 2001)
- **Transport modes:** the various networks used to transport people and/or goods; include **Motorized** and **Non-Motorized Transport (NMT)** modes.

Transport modes (as listed) L2·s4–9

Mode	Items on slide
Railways (s4)	Elevated railways; Underground (Tube) railways; Surface railways; Light Rail Transit (Trams)
Road transport (s5)	Motorcycles; Public transport; Non-motorized modes (walking & bicycles); Private cars; Heavy vehicles
Air transport (s6)	Helicopters; Hot air balloons; Commercial aircrafts; Private jets; Airships or blimps
Water transport (s7)	Inland Water Transport; International Water Transport; Ferry boats; Submarines; Cargo ships; Small boats; Cruise ships
Pipelines (s8)	e.g. pipelines carrying water from a reservoir to destination
Ropeways (s9)	Monocable; Bicable; Tricable ropeway

Transportation infrastructure L2·s10–17

Type	Content
Railway (s10)	Infrastructure for all types of railways incl. subways and elevated cable cars; railway lines, other structures, buildings, equipment, **with land on railway premises**. Pictures: tracks, tunnels & bridges, signages, crossings.
Road (s11)	Paved and unpaved roads + associated physical assets: signages, traffic lights, street lights, pedestrian walkways, earth works (speed breakers, mile stones), drainage works, culverts, bridges, buildings; bridges & tunnels.
Waterways (s12)	Navigable waterways (canals, rivers); **Navigation buoys** (inland waterways); **Light houses**; **Navigable aqueducts**; **Canal tunnels**. (Panama Canal working shown.)
Air traffic (s13)	Air routes and traffic control facilities; air navigation systems; air traffic flow management facilities. Pictures: ATC tower, control room, aircraft hanger, apron.
Terminals & stations (s14)	Terminals/waiting stations for passenger boarding and arrival for buses, railways, waterways, airways. **Passenger comfort, convenience and service is of prime importance.**
Ports/Seaports (s15)	Docking facilities and facilities for transfer of people and goods (container loading, port cranes).
Cycling (s16)	Bike paths, parking facilities.
Living streets (s17)	Designed for multiple uses, **restricting speed of vehicles**, pedestrian importance, incorporating children, recreation and green spaces.

Factors influencing EFFICIENCY of a transport system — 11 factors L2·s18–22 (L. R. Kadiyali 2017)

Speed · Safety · Adequacy · Frequency · Regularity · Integration · Responsibilities · Comfort · Cheapness · Fuel efficiency · Employment Generation

Factor	Meaning on slide
Speed	Essence of transport; faster transport reduces overall cost, reduces storage and inventorying; faster turn-around.
Safety	Safety of passengers and goods; highest possible standard of construction, maintenance and operation; protection against theft, fire and damage in case of goods.
Adequacy	Represents the **capacity** of a transport mode; should have sufficient capacity.

Frequency	Important characteristic; related to capacity consolidation and storage; assured frequency attracts people and goods.
Regularity	**Punctuality** and ability to adhere to scheduled service; gives confidence to clients.
Integration	Where mode requires transfer from one mode to another; **a system with few transfers** = efficiently integrated.
Responsibility	Ability to meet **liabilities for loss or damage**.
Comfort	Essential factor for **passenger** transport.
Cheapness	Main consideration for consumer preference; depends on total facility resources cost incl. construction, maintenance and operational costs.
Fuel efficiency	Global energy crisis demands consumers adopt fuel-efficient modes.
Employment generation	Any mode with good employment potential should be favoured.

Factors influencing DEVELOPMENT of a transport system L2·s23–27 (Rodrigue 2020)

Five factor groups: **Environmental, Historical, Technological, Political, Economic**. Four scales: **Local, Regional, National/Transnational, Global**.

Scale	Environmental	Historical	Technological	Political	Economic
Local (s24)	Hydrography & geomorphology	Cultural and settlement patterns	Roads	Zoning	Employment and distribution
Regional (s25)	Climate	Urban system	Railways and canals	Taxation and regulations	Modal competition and complementarity
National/Transnational (s26)	Distance	Nation state / colonialism / imperialism	Corridors and sea routes	Trade agreements	Markets
Global (s27)	Oceanic masses	Globalization	Air transport and telecommunications	Multilateral agreements (WTO)	Interdependency and comparative advantages

Transportation system components (Sussman 2006) L2·s28–32

Diagram (s28): Transportation System at centre; external perspective: **Government, General Public, Competition, Customer, Supply Industry, Financial Community, Stakeholders**.

Component	Items
Government (s29)	Taxes; Safety regulations; User fees; Financial regulation; Entry and exit from market; Monopolies; Provider of infrastructure; Provider of funding
Competition (s30)	Intra-modal; Inter-modal (Rail vs Truck); for the consumer's money (buy a two wheeler or take a vacation); Transportation/Communication system competition. **Common error: viewing the competition as static.**
Supply industry (s31)	Vehicle providers; Infrastructure builders (low-tech, high-tech); Components (electronics, materials); Research community; Insurance industry
Stakeholders (s32)	People/organizations who are **not customers or suppliers** but concerned: Environmental community; Abutters; The Public (quality of life, national defense, economic development).
Financial community (s32)	Provider of **Equity** and **Debt**

Multimodal transport L2·s33–35

- Combining multiple modes to transport people/goods effectively from source to destination; also called **"Combined transport"** or **"Intermodal transport"**.
- Ensures seamless travel by: **Vehicle mode integration** (car, BRTS, metro etc.); **Service integration** (private & public modes); **Infrastructure integration** (bus & metro stations, parking facilities etc.).

- Concept (s34): Home → **Access modes** (walking, bicycle/rickshaws, 2 wheeler, bus) → transfer stations (Car/Bus/Metro/BRTS/MRTS) → **Egress modes** (walking, bicycle/rickshaws, 2 wheeler, bus) → Office.
- **Challenges (s35):** poor infrastructure facilities; intermodal linkage at regional and local levels; current aged transport modes; enhance ICT; modernization of public passenger terminals; role of Government to facilitate MMT.

Conclusions L2-s36

- Transportation systems include all modes (**road, rail, air, waterway, pipeline and cable**) with infrastructure.
- Efficient system = efficient policy measures + good infrastructure + integrated modes + integration with ICT to tackle congestion, pollution, negative externalities. Sustainable solutions = the future.

Important Distinctions to Remember

- **Efficiency** factors (11, Kadiyali) vs **Development** factors (5 groups × 4 scales, Rodrigue) vs **Components** (Sussman).
- **Frequency** (how often) vs **Regularity** (punctuality/adhering to schedule) vs **Adequacy** (capacity).
- **Stakeholders** are NOT customers or suppliers; **Financial community** = equity + debt.
- Competition: intra-modal vs inter-modal (rail vs truck).

Image-only slides of this lecture were rendered and read; their content is in **Part 2A (end of Part 2) under Lecture 2**.

Lecture 3 — Concept of Sustainability (45 slides)

Why present development may be unsustainable L3·s3–6

- Challenges since the **Industrial Revolution**; industrialization contributed to population growth, employment generation, wealth creation. Pollution = essential by-product of the present **fossil-fuel based** economic system. Some thought leaders: development is at the cost of future generations → unsustainable.
- **Factors contributing to unsustainable development (s4):** polluting industrialization; increasing economy & per capita income with huge gap between rich and poor; unchecked population growth; continuous depletion of non-renewable resources; climate change.
- **Negative externalities (s5), 14 listed:** air pollution; noise pollution; accidents; global warming; depletion of energy resources; consumption of land; loss of habitat; hazardous materials; vibration; visual intrusion; waste disposal; water pollution; congestion; social disruption.
- **Impacts (s6):** food shortage; ecological crisis; slowing economic growth; raising social unrest; energy crisis; environmental pollution.

Definitions L3·s7–10

- **Sustainable development (Brundtland report, 1987):** "development that meets the needs of the present without compromising the ability of future generations to meet their own needs."
- **Development (Per Becker, 2014):** preferred expected scenario of change in a set of variables over time, from a current to a desired state, including purposeful activity.
- Negative events/processes may cause **deviations** from the preferred scenario. **Sustainable development** = development that can be **maintained over time** and **safeguarded** from impact of negative events.
- Anticipating potential deviations (**risk scenarios**) is a **requisite** for sustainable development.
- **Evolution of the term (s11):** Embryonic period (**before 1972**); Molding period (**1972–1987**); Developing period (**after 1987**).

Three pillars (IUCN 2006) L3·s12–15

Social · Environmental · Economic sustainability. Criteria (EPA):

Pillar	Criteria and meaning
Social (s13)	**Environmental justice**: protect health of communities over-burdened by pollution by empowering them to act. **Resource security**: protect, maintain, restore access to basic resources (food, land, energy). **Human health**: protect, sustain, improve. **Participation**: open and transparent processes engaging relevant stakeholders. **Education**: enhance education about sustainability. **Sustainable communities**: promote development, planning, building or modification of communities to promote sustainable living.
Environmental (s14)	**Ecosystem services**: protect, sustain, restore critical natural habitats and ecosystems. **Green engineering and chemistry**: develop chemical products/processes to reduce/prevent hazards, reuse/recycle chemicals, treat, dispose properly. **Resource integrity**: reduce adverse effects by reducing waste generation, increased recycling, proper waste management; restore resources by mitigating/cleaning accidental or intentional releases. **Environmental stressors**: reduce effects of stressors (pollutants, GHG emissions, genetically modified organisms). **Air quality**: attain and maintain air-quality standards, reduce risk from toxic air pollutants. **Water quality**: reduce exposure to contaminants in water bodies.
Economic (s15)	**Jobs**: create or maintain job opportunities (e.g. green jobs). **Incentives**: encourage sustainable practices (e.g. sustainable logging). **Prices**: cost structure that accounts for externalities to production processes (positively impact costs of services/products). **Natural resource accounting**: incorporate natural capital depreciation in accounting indices and ecosystem services in cost-benefit analysis (e.g. **Green net national product**). **Supply and demand**: price/quantity changes that alter economic growth, environmental health and social prosperity (e.g. increasing supply of green energy sources to reduce need for fossil fuels).

- **Principles (s17):** sustainable development should be **environmentally bearable, socially equitable, economically viable**.
- **Basic principles (s18, Justice Mensah 2019):** minimizing use of non-renewable resources; minimizing impacts on natural environment; protecting biodiversity; using renewable resources in a sustainable manner.
- s16: Economy as subsystem of global ecological system — inputs: solar energy (Sun), non-renewable and renewable resources; outputs: waste, recycling, waste heat, damages (diagram).

Sustainable transport L3·s20–29

- **Definition (UNDESA 2014):** provision of services and infrastructure for the mobility of people and goods, advancing economic and social development to benefit today's and future generations in a manner that is **safe, affordable, accessible, efficient, and resilient**, while **minimizing carbon and other emissions and environmental impacts**.
- s21 figure words: Some, Appropriate, Measure, Sustainability, Resources, Technology, Needs, Environment, Ecology, Time.
- **Goals by dimension (s23):** Social → **Reduce congestion, Enhance safety**; Economic → **Expand economic opportunity, Increase value of transportation assets**; Environmental → **Improve air quality**.

Sustainable transport is… (s24–25)	Linked SDG
Safe	Good health and wellbeing (**SDG 3**)
Affordable	Zero Hunger (**SDG 2**)
Accessible / Efficient	Affordable and Clean Energy (**SDG 7**); Responsible consumption and production (**SDG 12**) [as grouped on slide]
Resilient	Industry Innovation and Infrastructure (**SDG 9**)
Minimizing carbon & other emissions and environmental impacts	Climate Action (**SDG 13**); Sustainable cities and communities (**SDG 11**)

[Slide 24 lists the attributes and SDGs in two columns; exact one-to-one pairing for "Affordable/Accessible/Efficient" is not clear from extracted text.]

- **Objectives (s26):** "**5 Is and 5 Cs**" approach (UNDESA 2014) — details are in an image (not extractable).
- s27–29: Goals, Objectives & Indicators for Economic / Social (Todd Litman 2021) / Environmental sustainability — image tables (not extractable).
- **Why emphasis now? (s30):** growing awareness of problems; increasing intolerance with use of "free goods or services"; mistrust of global organizations; information age has increased power of third parties; expectation that something be done.

Examples L3·s31–38

Example	Facts
Octopus Smart Card, **Hong Kong** (s31)	**93%** of transport trips by public transport (**48%**) or walking (**45%**). Contactless smart card payment system; started as public-transport fare collection in **1997**; also used for access control (residential/commercial buildings), integrated public transport modes, payment/settlement, schools.
Reduce, Recycle, Reuse — **Grey County, Ontario** (s32)	Issue: thousands of used car tires. Solution: facility to recycle tires; rubber used in production of **asphalt pavements** within the county.
Last Mile Challenge — **Chile** (s33)	Inclusive transport for low-density rural areas; "**Leave no one behind**" (2030 Agenda). Connect rural areas disconnected from roads/rail/public transport. Ex: inclusive **waterborne** transport in extremely low-populated southern Chile; **free bus transport for handicapped children** from remote areas.
Dedicated Freight Corridor (DFC), **India** (s34)	One of the largest transport infrastructure projects in India; increases relative share of **rail** (energy efficient, environment friendly, less carbon-intensive). Expected to save **>450 million tons of CO_2 in first 30 years** of operation.
Eastern Canada **RWIS** (Road Weather Information Systems) (s35)	Economically beneficial via **reduced road salt use**; increases driver safety (improved salt management); lessens impact on roadside vegetation and wildlife.
BRT, **Johannesburg** (s36)	Impetus: hosting **2010 FIFA World Cup**. **First full BRT system in Africa**. Total cost **US$749 million**; economic returns estimated (2026) **US$900 million** [year as written].
Copenhagen (s37)	Aims to be "eco-metropolis", world's best city for cyclists, **CO_2-neutral by 2025**. **96%** can walk to a large green/blue recreational area in **<15 min**. **>50%** cycle to workplace/education. Noise-reducing asphalt, traffic planning, speed reduction reduced traffic noise.
Aggregate recycling (s38)	See table below. RAP = recycled asphalt pavement.

Material	Sweden	Germany	United States
Blast furnace slag	45%	100%	90%

Steel slag	100%	92%	—
Coal bottom ash	—	97%	31%
Coal fly ash	—	88%	27%
RAP	95%	55%	80%

[Slide lists Sweden: slag 45, steel slag 100, RAP 95; Germany: 100, 92, 97, 88, 55; US: 90, 31, 27, 80 — column assignment of US items by order on slide.]

- s39–42: Highway 59 good day vs bad day (Houston flood, June 2001 photos); Los Angeles smoggy vs clear day — "Los Angeles used to be the **most polluted air basin in the United States**".
- **Conclusion (s43):** negative externalities of non-renewable resources; global warming & climate change; increasing awareness; fossil-fuel economy vs renewable-energy-centric economy; journey towards sustainability — "Are we doing our share in the transportation sector?"

Important Distinctions to Remember

- Pillars: **Social, Environmental, Economic**. Principles: **environmentally bearable, socially equitable, economically viable**.
- Brundtland (**1987**) definition vs Per Becker "development" definition (change in variables from current to desired state).
- Hong Kong: 93% = 48% PT + 45% walking. Johannesburg = first full BRT **in Africa**. DFC = rail share ↑, 450 million tons CO_2 / 30 years.
- Economic criteria example: **Green net national product** (natural resource accounting).

Image-only slides of this lecture were rendered and read; their content is in **Part 2A (end of Part 2) under Lecture 3**.

Lecture 4 — Current Scenario of Transportation in India (44 slides)

Urban population & problems L4·s3–13

- **2011 census: 31.2%** of India's population (**377 million**) urban. UN estimates: **40% (590 million) by 2030**; **58% (875 million) by 2050**. About **63% of GDP** contributed by the ~30% urban population. Scope for **PURA** (Shri APJ Abdul Kalam).
- **Transport problems (s4):** growth in private vehicles; severe congestion; increasing GHG emissions; increasing road accidents; deteriorating air quality.
- **Private vehicle growth (s5):** average annual growth rate **10%**; private vehicle ownership increased to **438% in 2019** vs base year **2001**.
- **Share of motorized vehicles (2011) (s6):** two-wheelers have most share (**about 75%**) followed by private cars; two-wheeler ownership **doubled in 2020** vs 2011.
- **Registered motor vehicles (s7, MORTH 2016):** total increased from **0.3 million (1951)** to **230.03 million (2016)**.

Type	1951	2000	2016
2 wheelers	27,000	34.12 million	169 million
Cars, jeeps & taxis	159,000	6.1 million	30 million
Buses	34,000	0.5 million	1.8 million
Goods vehicles	82,000	2.7 million	10 million
Others	4,000	5.3 million	18.5 million

- **Delhi (s9):** motor vehicle use doubled between **2010 and 2020**.
- **Infrastructure gap (s10):** registered vehicle increase in last decade **219%**; urban road infrastructure increase **124%**. Rapid motorisation → severe congestion, longer journeys, higher per capita trips.
- **Congestion (s11): 3 Indian cities in top 10** congested cities: **Mumbai 2nd, Bengaluru 6th, Delhi 8th**, Pune **16th** among **416 cities in 56 countries**.
- **Road accidents (s12):** over **151,000** fatalities annually ≈ **414 each day**; India ≈ **6%** of global road traffic incidents; almost **70%** involve young Indians.
- **Potholes (s13):** over **4770** accidents in 2019; accidents decreased but deaths due to potholes increased significantly in 2019.

Air pollution / GHG L4·s14–20

- WHO 2018: **4.2 million** deaths/year worldwide from ambient air pollution. 2017 Global Burden of Disease: ~**1.1 million** premature deaths/year in India; **fifth leading cause of death**.
- **GHG by sector in India (WRI CAIT 2018, s15):** Energy **68.7%**; Agriculture **19.6%**; Industries **6%**; Land-use change & forestry **3.8%**; Waste **1.9%**.
- Road transportation > **90%** of GHG from transportation sector in India (2014, MoEFCC) (s16).
- G20 (s19): **Turkey** transport emissions per capita increased **38%** (2013–2018); **India and China 2nd and 3rd largest contributors**; **Saudi Arabia** reduced by **25%** (2013–2018).
- Transportation sources ≈ **one third of PM pollution** in India (s20).
- s17 (GHG from transport worldwide 2018) and s18 (per-capita emissions G20 excl. aviation, 2018) are charts (image).

Public transport & policy L4·s21–29

- Public transport share only **18.1%** of total work trips; 2015: **35 million** daily motorcycle commuting trips (excl. personal); more people use motorcycles than bus (**22.9 million**) (s21).
- **National Urban Transport Policy (NUTP), 2006** — GOI policy framework for sustainable mobility (s22).
- **Smart cities (s23):** two flagship programmes in **2015**: **100 Smart Cities** and **AMRUT** (Atal Mission for Rejuvenation and Urban Transformation) for **500 cities** with population ≥ 100,000; funding **$8 billion** and **$8.3 billion** respectively. Smart cities focus on **core infrastructure service**; AMRUT adopts a **project approach** for basic infrastructure services.
- **Metro (s25):** Delhi Metro = largest, **213 km**, **2.8 million** passengers/day. MoUD: ~**316 km** in operation; >**500 km** under construction.

- **City bus (s26–27):** >**1.6 million** buses registered; public bus sector operates **170,000** buses carrying ~**70 million** people/day. **NURM launched 2009**. BRT: ~**222 km** operational by 2015, remaining **282 km** under construction; MoUD financial assistance to **11 cities** for **504 km** BRTS.
- **Future bus (s28–29):** full indigenous retro-fitted electric buses with hybrid technology; hindrance = financial implication/capital cost; hybrid bus **INR 23 million** vs premium diesel bus **INR 9 million**. **BMTC** (Bengaluru) = first in India to launch an **open-loop EMV contactless smartcard** (not integrated with other modes). **MMRDA** appointed London's **TfL** to prepare integrated ticketing plan for Mumbai Metropolitan Region.

Non-motorized transport (NMT) L4·s30–35

- NMT in India = **walking, cycling, cycle rickshaws**. Green mode: low carbon footprint, minimal energy consumption, zero local emissions, inexpensive, health benefits; low-income households largely depend on NMT; still neglected in favour of capital- and infrastructure-intensive modes.
- NMT share (walking+cycling) **40–60%** in 1980s; declining since 1980s except e.g. **Chennai and Patna** (s31).
- Bicycle trips **10–30%** of total trips; declined **22% in Kanpur** by 1995, compared to **58% in Delhi** and **86% in Jaipur**; Patna increased **83%** between 1990s and 2000s and again by **6%** by 2008 (s32). *[Slide numbers 83/86/58/58 appear scrambled in the graph; values as read.]*
- NMT blamed for congestion; rickshaws considerably increase road capacity on segregated lanes; **24%** of Delhi metro trips depend on rickshaws as feeder (s33).
- **MoUD policy (s34–35):** NUTP vision: NMT = **last-mile connector** and for short independent trips. First two principles of National Mission for Sustainable Habitat (PM's National Action Plan on Climate Change) = **"walk" and "cycle"**. MoUD developed **Service Level Benchmarks (SLBs)**. **JNNURM 2005** required NMT as integral component. 2006 NUTP emphasised safe NMT infrastructure. **BUT: NO DIRECT PROJECTS aimed exclusively at NMTs in India.**

Public transport challenges & post-COVID L4·s36–42

- s36 Fund allocation / consequences of investments in roads; s37 Indian cities **lag behind** in physical coverage of public transport network vs cities across globe; s38 cost & capacity of mass transit — figures (images).
- s39–41: Estimated modal share pre/post COVID-19 (TERI survey, **April 2020**); distance to work and age vs post-COVID mode choice of initial public transport users — charts (images).
- **Conclusion (s42):** shift **from** public transport expected post-COVID; subsequent increase in private vehicles; increase in NMT, especially cycling for short-distance trips; scope of NMT as major short-distance choice; sustainable measures in India still at **policy level**.

Important Distinctions to Remember

- **Smart Cities (100, core infrastructure)** vs **AMRUT (500 cities ≥100,000 pop., project approach)**; funding $8 bn vs $8.3 bn.
- **NUTP 2006** vs **NURM 2009** vs **JNNURM 2005**.
- Vehicles +219% vs urban roads +124% (last decade). Private vehicles +438% in 2019 vs 2001.
- Two-wheelers ≈75% share; more motorcycle commuters (35 m) than bus (22.9 m).
- Energy sector = 68.7% of India GHG; road = >90% of transport GHG.

Image-only slides of this lecture were rendered and read; their content is in **Part 2A (end of Part 2) under Lecture 4**.

Lecture 5 — Climate Change: Indicators and Impacts (44 slides)

Basics L5·s3–8

- **Climate change:** change in usual weather found in a place (e.g. rain per year, usual temperature for a month/season); also change in Earth's climate (usual temperature; places where rain and snow fall). **Weather can change in a few hours; climate takes hundreds or even millions of years to change.** (NASA 2014)
- **Causes (s4):** Natural (**not controllable**): Earth's distance from the sun; oceans, volcano eruptions etc. Humans (**controllable**): anthropogenic activities; GHG emissions.
- **Is warming manmade? (s5):** solar energy received followed the Sun's natural **11-year cycle** with **no net increase since the 1950s**, while global temperature rose markedly → "extremely unlikely that the Sun has caused" the warming trend of past half-century.
- s6: yearly temperature vs twentieth-century average 1880–2019 (red/blue bars) and atmospheric CO_2 1880–2019 (grey line). s7: average global temperature & world carbon emissions from fossil fuel burning (millions of tons) 1800–2014.
- **Warmest years (s8): 2020 = 2nd warmest** on record; **2016 warmest**; 2019 3rd warmest; world's **7 warmest years all since 2014**; **10 warmest all since 2005**.

Climate change drivers: GHGs L5·s9–13

- GHGs = **most significant driver** of observed climate change since the **mid-20th century**. Human-activity GHG sources: electricity, transportation, agriculture, commercial & residential.
- **Global warming potential (GWP):** concept used to convert amounts of other gases into **carbon dioxide equivalents**.
- **Radiative forcing:** the heating effect caused by greenhouse gases in the atmosphere (s12).
- s11: global atmospheric concentrations of CO_2, CH_4, N_2O & O_3 1950–2015 (chart). s13: radiative forcing by human activities (chart).

GHG	Sources (s10)	Avg. lifetime	100-yr GWP
Carbon dioxide	Burning of fossil fuels, solid waste, trees and wood products; deforestation and soil degradation	Varies*	1
Methane	Production and transport of oil, natural gas & coal; livestock and agriculture; waste decomposition	12.4 years**	28–36
Nitrous oxide	Agricultural and industrial activities; combustion of fossil fuels and solid waste	121 years**	265–298
Fluorinated gases	Industrial processes and commercial and household uses	Few weeks to 1000s of years (varies by gas)	Varies upon the gas

*CO_2 may get absorbed quickly (ocean surface) or may remain even thousands of years. **Perturbation lifetimes. (Table verified against the rendered slide.)

Climate change indicators L5·s14–26

Indicator	Facts
Weather & climate (s14–15)	Rising global average temperature → widespread changes in weather patterns. Extreme events (heat waves, large storms) likely more frequent/intense. Extreme variation = threat (illness to death). **Positive:** increased precipitation can replenish water supplies and support agriculture. **Negative:** intense storms damage property, loss of life, displacement, disrupt transportation, telecommunications, energy, water supplies.
Oceans (s16–17)	Cover about **70%** of Earth's surface. As GHGs absorb more energy, oceans absorb more heat → higher sea surface temperature and rising sea level. **Positive:** oceans reduce climate change by storing large amounts of CO_2. **Negative:** dissolved carbon makes seawater more **acidic**; may alter biodiversity and productivity; corals and shellfish find it harder to build skeletons and shells.
Ocean acidity (s18)	Ocean CO_2 levels rose with atmospheric CO_2 → increase in acidity (a **decrease in pH**).
Ocean heat (s19)	Ocean heat content 1955–2015; measured in **joules**; compared with **1971–2000 average (set to zero)**.

Sea surface temperature (s20)	Increased during 20th century; **1901–2015**: rose at average **0.13 °F per decade**.
Sea level (s21)	Rose throughout 20th century, rate accelerated. Absolute sea level rose avg **0.06 in/yr** (1880–2013). Since **1993**: **0.11–0.14 in/yr** — roughly **twice** as fast as long-term trend.
Snow & ice (s22)	Forms: sea, lake, river ice; snow cover; glaciers, ice caps, ice sheets; frozen ground. Altered in response to relatively minor temperature changes; reduced snow/ice influences air temperatures, sea level, ocean currents, storm patterns.
Arctic sea ice (s23–25)	Dwindling; March = maximum, September = minimum monthly extent (1979–2016); melt season length has **increased** over the years (1979–2015); age of ice at minimum September week 1983–2015 (charts).
Glaciers (s26)	Losing mass worldwide since at least the **1970s**, contributing significantly to sea level rise; rate appears to have **accelerated over roughly the last decade**.

Impacts L5·s27–41

- **Health & society (s27–28):** affects public health, agriculture, water supplies, energy, land use, recreation. Health impacts are **complex, mostly indirect**; health-related climate indicators are challenging and still emerging. Warmer temperatures → hotter days, longer heat waves → more heat-related illness and death; storms → flooding, high winds; reduced air quality, spreading diseases.
- **Chicago heat wave, Cook County, July 11–27, 1995 (s29):** may have been hundreds more deaths than reported as "heat-related" on death certificates.
- **Lyme disease (bacterial) (s30):** US incidence approx. doubled 1991→2014: **3.74** → **7.95** cases per 100,000.
- **West Nile virus (s31):** neuroinvasive cases occur in **<1%** of infected people; affect brain / neurologic dysfunction (data 2002–2014).
- **Ecosystems (s32–35):** provide food, clean water, services; climate affects water cycle, habitats, animal behaviour (nesting, migration), timing (flower blooms); risk of harm/extinction. **Wildfires:** disrupt ecosystems, damage property, risk to people, air pollution far from source. **Streamflow:** temperature, precipitation, snowpack, glaciers affect rate and timing of peak flow. **Bird wintering ranges:** North American birds shifted wintering grounds **northward by avg >40 miles since 1966**; several species by hundreds of miles (1966–2013).
- **Disasters 1998–2017 (s37–39, UNISDR):** climate-related disasters are more significant; climate-related deaths are more in the last decade (charts).
- **COVID-19 (s40, s42):** CO_2 emission fall is **only temporary**; "Climate change is not on pause."

Key facts (WHO 2018) L5·s41

- Climate change affects social & environmental determinants of health: **clean air, safe drinking water, sufficient food, secure shelter**.
- **2030–2050:** ~**250,000** additional deaths/year from **malnutrition, malaria, diarrhoea, heat stress**.
- Direct damage costs to health: **USD 2–4 billion/year by 2030**.
- Reducing GHG via better transport, food, energy-use choices → improved health, particularly through reduced air pollution.
- **Conclusion (s42):** climate change affects every country on every continent; sustainable development (essentially incl. **Sustainable Transportation Systems**) is the **most feasible mitigation measure**.

Important Distinctions to Remember

- **Weather** (hours) vs **climate** (hundreds to millions of years). Natural (not controllable) vs human (controllable) causes.
- Oceans: positive (store CO_2) vs negative (acidification, lower pH).
- SST rise **0.13 °F/decade** (1901–2015) vs sea level **0.06 in/yr** (1880–2013) vs **0.11–0.14 in/yr** (since 1993).
- Lyme disease: bacterial; 3.74→7.95 per 100,000. West Nile neuroinvasive: <1%.
- Warmest: **2016** (1st), **2020** (2nd), 2019 (3rd).

Image-only slides of this lecture were rendered and read; their content is in **Part 2A (end of Part 2) under Lecture 5**.

Lecture 6 — Impacts of Transportation Systems-I (30 slides)

Transport–environment link L6·s3–7

- Transportation and environment = **paradoxical**: substantial socio-economic benefits but impacts environmental systems ("paradox of mobility and its costs").
- **Level of contribution:** transport contributes (with other anthropogenic and natural causes) **directly, indirectly, and cumulatively**.
- **Scale of impact: local** (noise and CO emissions) → **global** (climate change); also continental/national/regional (**smog and acid rain**).
- **Activities affecting the environment (s6):** Infrastructure (resources, energy, land); Vehicle manufacture (resources & energy consumed); Vehicle operation (fossil fuel consumption); Vehicle maintenance (consumption & disposal of parts, lubricants); Vehicle disposal.

Energy L6·s8–13

- Transport = about **20–25%** of world energy use; strong correlation with level of development; **87%** of world's total power production from fossil fuels.
- **World energy consumption (s10):** Coal **39% (1965)** → **28% (2016)**; Oil **40% (1965)** → **33% (2016)**; Natural gas **15% (1965)** → **24% (2016)**; Renewables **3% (2016)**.
- **Transport sector energy share (OECD, s11): 24.2% (1971)** → **33.7% (2016)**.
- **Four major factors of energy use by transportation (s12):** Vehicles (fuel efficiency); Infrastructure (transport efficiency); Demand (economic activity); Spatial infrastructure (average transport distance).
- **Fuel type (s13):** oil-related fuels = **94%**. Gasoline **43.5%** most used; diesel **34.5%**; heavy oil, biofuels, kerosene **21.8%** [as written]; electricity and gases (e.g. LNG).

Emissions L6·s14–17

- Energy generation = most significant contributor to global GHG; **transport = 22% of global GHG emissions**.
- CO_2 sharp increase **after 1960s**; **China and India** became the world's largest carbon emitters in early 21st century.
- **Global mean sea level (s17):** main factors: increase in quantity of water in oceans; increase in global average temperatures. Increased risk of flooding & damage to coastal infrastructure.

Air quality & health L6·s19–22

- Transport linked to climate change and air quality due to growing motorization and congestion.
- Sources: highway vehicles, marine engines, locomotives, aircraft (gas + particulate emissions). Common pollutants: **lead (Pb), carbon monoxide (CO), nitrogen oxides (NOx), silicon tetrafluoride (SF_6) [as written], benzene and volatile components (BTX), heavy metals (zinc, chrome, copper, cadmium), particulate matter (ash, dust)**.
- **Impact types (s22): Direct** (noise pollution; CO emission); **Indirect (secondary/tertiary)** – higher consequence than direct (e.g. land clearing; particulates from incomplete combustion; respiratory & cardiovascular problems); **Cumulative** – additive, multiplicative or synergic (e.g. climate change).

Transport ↔ climate L6·s23–27

- Fossil-fuel vehicles (cars, buses, 2 & 3 wheelers, steam trains, boats) → CO_2 → greenhouse effect. Earth's average temperature up **0.8 °C** since early 20th century [slide: "0.80C"].
- s24: impacts of climate change on transport operations and infrastructure (figure). s25: carbon cycle – carbon pools in GtC; red = human emissions (figure).
- **Emissions by mode (s26):** walking and cycling = best; sharing a vehicle reduces emissions vs single trips; mass/public transport has fewer emissions per trip.
- **Environmental Kuznets Curve (EKC) (s27):** early stages of economic growth → pollution and degradation **increase**; beyond some level of income per capita the trend **reverses** → at high income, growth leads to environmental improvement (example fig: sulfur emissions).

Conclusion — GHG reduction measures L6·s28

- Reduced use of private automobiles · increased use of public transport · alternate clean fuels · promotion of NMT · integrated transport development.

Important Distinctions to Remember

- Impact types: **direct** vs **indirect** (higher consequence) vs **cumulative** (additive/multiplicative/synergic).
- Scale: local (noise, CO) vs regional (smog, acid rain) vs global (climate change).
- Transport = 20–25% of *world energy use* vs 22% of *global GHG* vs 33.7% of *OECD energy (2016)*.

Image-only slides of this lecture were rendered and read; their content is in **Part 2A (end of Part 2) under Lecture 6**.

Lecture 7 — Impacts of Transportation Systems-II: Human Health (49 slides)

Transport and health L7·s3–10

- Transport negatively impacts health in **three broad ways**: **Traffic accidents; Environmental pollution; Spread of infectious diseases**. (K. J. Button 2010)
- **1989:** International Agency for Research on Cancer concluded diesel and gasoline engine exhausts are **probably carcinogenic** to humans. **1998:** California Air Resources Board identified **diesel PM** as a **toxic air contaminant (TAC)**. Children near heavily travelled roads: ~**50% higher risk** of respiratory symptoms from ambient PM10.
- Private motorcars → reduced physical activity; lack of physical activity = major risk for **coronary heart disease** (leading cause of mortality in Europe). Walking & cycling: promote health; decrease noise and air pollution.
- s6 framework (Khreis 2017): Urban transport policy & planning (built environment; transport provision & mode choice) → motor vehicle crashes, air pollution, noise, heat islands, lack of green space and biodiversity loss, physical inactivity, social exclusion, community severance, climate change → premature mortality / morbidity.
- **Mental health (s8–10):** improving walking, cycling, public transit improves mental health via **increased physical activity** and **community cohesion**. Increased walkability **reduces symptoms of depression in older men** and **frequency of dementia in women and men**. High-quality public transport reduces commute stress vs driving. Children able to play unhindered by traffic found **socially active twice** as much; lack hinders independence, linked to depression. Effects: risk-taking and aggressive behaviours; depression; post-traumatic effects of accidents; high traffic → social isolation (higher mortality and morbidity in elderly).

Accidents L7·s11–12

- Globally every year >**1.17 million** die in road crashes; >**10 million** crippled or injured. ~**70%** involve young people. India ≈ **6%** of global incidents. ~**45,000** deaths in the US in 2006.
- Among developed countries, per capita traffic fatalities **increase with** per capita vehicle travel; **United States has highest per capita crash rate** (OECD).

Traffic-generated air pollution L7·s13–16, 27–31

- Motor vehicle traffic = main source of ground-level urban concentrations of hazardous air pollutants. **CO and benzene inside cars ≈ 2–5 times higher than at roadside.** Car users more exposed than pedestrians, cyclists, public transport users.
- **Main pollutants:** PM (PM10, PM2.5); CO; NOx; NMVOCs; O_3; benzene. **Trace:** CH_4, CO_2, SOx, TSPs.
- **Short-term PM10 effects:** increased mortality; respiratory & cardiovascular diseases; more respiratory symptoms and medication use by asthmatics; reduced lung function. Recurrent cumulative exposure ↑ morbidity, ↓ life expectancy.
- **Determinants (s27):** number of vehicles; age of vehicles; engine type & condition; type of fuel; meteorology; shape of urban environment; traffic organization.
- Heavy vehicles = major polluters: diesel truck produces **50–100 times** more fine and ultra-fine particles/km than a car with catalytic converter. Modern diesel emits less PM2.5 but **more ultra-fine particles**; ultra-fine particles (number rather than mass) suspected main cause of health effects → new diesels may cause more harm.
- **Health end-points (s29):** mortality (long term); cardiovascular; respiratory; acute bronchitis (<15 y); asthma exacerbation (<15 y and ≥15 y); restricted physical activity (15–64 y); lung cancer.
- Fine particulate pollution responsible for most of the **3–4 million** annual air-pollution deaths worldwide (s31).

Los Angeles case L7·s17–26

- People in LA county exposed to **60% more** vehicular pollution than state average. Thick smog from the **1940s**, lasted over **3 decades**; LA = "City of Stars" also "City of Smog"; 1964/1966 photos (gas masks, low visibility, accidents). **Clean Air Act passed 1970** → harsher control measures. LA used to be the most polluted air basin in US.

Health improvement strategies L7·s32–43

Strategy	Key facts

WHO **DPSEEA** framework (s32)	Driving force, Pressure, State, Exposure, Effect and Action.
Policy implications (s33–34)	Unleaded petrol & catalytic converters helped. **Holistic approach**; tackling single pollutants in isolation can be counterproductive (e.g. promoting new diesel to cut CO_2 → more ultra-fine particles). **Co-benefits:** consider the pollution mix; prioritise GHG actions that also reduce other pollutants like PM.
Traffic calming & speed control (s35)	Reduces crash frequency & severity; reduces total vehicle travel. Per-km emissions decline from high to moderate speeds but **increase at very low speeds (<20 km/h)** and stop-and-go (e.g. speed bumps).
Active transport (s36)	Better sidewalks, crosswalks, paths reduce crash risk; reduces total crash rates, air pollution, congestion; increases physical activity. **1% shift from motorized to NMT typically reduces emissions by 2–4%.**
Public transit improvements (s37)	More routes; longer operating hours; frequent services; quality vehicles & terminals; grade separations; real-time user information. Public transit travel has **lower traffic fatality rate** than other automobile travel.
Transport pricing reforms (s38)	One-cent US state gasoline tax increase → per capita traffic fatalities **−0.25%**, per vehicle-mile **−0.26%** (Grabowski & Morrisey 2006). Sustained **20%** gasoline price rise would prevent **2,000** crash deaths (5%) + **600** air pollution deaths (Leigh & Geraghty 2008). **1%** vehicle travel reduction reduces crashes by **more than 1%** (Chi et al. 2010).
Mobility management marketing (s39)	Programs encouraging shift from automobiles: commute trip reduction programs; transportation management associations; ride matching & vanpool support; direct marketing.
Smart growth (s40–43)	Compact & mixed development; connected path/road networks; transport–land-use integration; walkability; efficient parking management. Residents drive **20–40% less**; lower per capita casualty rates; more active transport. Caveat: compact development may increase some pollution exposure (**canyon effect**) → mitigate with cleaner transit vehicles & sidewalks away from traffic.

Case study: in-car aerosol exposure, 10 global cities L7·s44–46 (Kumar et al. 2021)

- PM2.5 collected every **1 minute** with portable laser particle counter from **backseat**. **3 car settings:** windows-open; windows-closed with fan on; windows-closed with recirculation. Morning peak, evening peak, off-peak.
- Hotspots: data points exceeding **90th percentile (P90)** of 1-min PM2.5 during window-open setting. Compared with GDP, value of statistical life (VSL), road density.
- Inference: Guangzhou (CAN) & Addis Ababa (ADD) hotspots = only ~**26–28%** of time but ~**54–56%** of inhaled dose. Hotspot length **increases with road density**; **negative correlation** with GDP and VSL (except Guangzhou).

Important Distinctions to Remember

- 3 broad health impacts: accidents / pollution / infectious diseases.
- IARC 1989 (probably carcinogenic) vs CARB 1998 (diesel PM = TAC).
- Car interior CO & benzene 2–5× roadside; truck vs catalytic car 50–100× fine/ultra-fine particles.
- 1% NMT shift → 2–4% emission cut; smart growth → drive 20–40% less; 20% gasoline price → 2,000 + 600 deaths.
- Speed: per-km emissions fall high→moderate, rise <20 km/h.

Image-only slides of this lecture were rendered and read; their content is in **Part 2A (end of Part 2) under Lecture 7**.

Lecture 8 — Impacts of Transportation Systems-III: Flora & Fauna (51 slides)

Classification of impacts L8·s3

Direct impacts	Indirect impacts
Loss of wildlife habitat · Habitat fragmentation · Fauna traffic mortality · Corridor restrictions (the barrier effect) · Disturbance and pollution · Traffic noise · Visual impacts · Interruption of biogeochemical cycle	Accessibility · Ecological disequilibrium · Contamination of the biota · Forest fires

Four main effects on wildlife L8·s5–6 (EEA 2011)

1. Habitat loss 2. Traffic mortality 3. Population subdivision 4. Barrier effect — reduce population size and increase risk of extinction. After a **time lag (order of decades)**: population size smaller, greater relative fluctuations over time, more vulnerable.

Direct impacts in detail

Impact	Key points
Loss of wildlife habitat (s7–9)	Natural habitat significantly altered; may seem negligible at larger scale but serious locally (depends on location of infrastructure, affected habitat & species). Infrastructure on **hilly areas** has greater consequences. With increasing infrastructure density, undisturbed habitat areas shrink and become inaccessible. Remnant fragments may become too small/isolated → local extinction. **Critical threshold of road density is species-specific**, also depends on landscape & infrastructure characteristics.
Habitat fragmentation / barrier effect (s10–13)	Roads/railways impermeable to animals. High-density traffic roads and **high-speed railways** prevent free passage; limits ability to search for food, shelter, mating. Physical barriers: completely fenced roads/railways; unsuitable surfaces or crossing objects (e.g. road verges). Road cut → sum of two parts < value of initial whole; fragments into weaker sub-units, more vulnerable to invasions and degradation. Theory (s13): increasing traffic increases road accidents **linearly**, until noise & vehicle movements repel animals from crossings.
Fauna traffic mortality (s14–16)	Most evident and well-known negative impact. Mostly affected: rare species moving long distances (e.g. **otter, large carnivores such as tigers**); seasonal migratory species such as **amphibians (frogs, toads)**. Concentration of casualties depends on: temperature; precipitation/time of day; ecological factors (breeding dispersal, food supply, seasonal migrations, age & sex); location/landscape, road specifications, traffic density, crop rotation. Examples shown: NH 67 (Gundlupet–Ooty); NH 212 (Mysore–Kozhikode).
Aquatic habitat damage (s17–19)	Erosion from poorly constructed/rehabilitated sites → downstream **siltation**, ruining fish spawning beds. Constricted channel flows at crossings make current too fast for some species. Altered flood cycles, tidal flows, water levels upset trophic dynamics (plankton life cycle → food chain). **Behavioural barriers:** mostly larger species; avoidance of areas near roads/railways, avoid crossing large open spaces.
Corridor restrictions (s20–22)	When a road blocks a wildlife corridor: (1) cessation of corridor use, (2) increased mortality by collisions, (3) delay in migration → weakening/disappearance of an entire generation. On busy roads death rate for amphibian/slow-moving populations can be as high as **one in ten**. Example: transport routes along Rajaji National Park, Uttarakhand.
Disturbance & pollution (s23–24)	Mountain environments more sensitive. Hydrological changes (cuttings, embankments → soil erosion); chemical pollution (oxides, hydrocarbons, particulates, heavy metals from exhausts); **de-icing salt** (sodium, chloride) in winter; herbicides in summer; gasoline/oil leaks in accidents. **Olivia oil spill 2011, Tristan da Cunha:** 40,170 GT bulk carrier grounded on Nightingale Island; leaked **1400 t Intermediate Fuel + 70 t Marine diesel**; oiled ~**3700** endangered Northern Rockhopper penguins.
Traffic noise (s25–27)	Rarely immediate physiological effect on humans; long exposure → psychological stress → physiological disorder. Wild animals treat noise as sign of humans → avoid (wild reindeer, elk, caribou, brown bear). **Birds very sensitive:** minimum density near roads; interferes with vocal communication, territorial behaviour, mating success; verge vegetation, adjacent habitat, relief alter noise. **Most vulnerable species:** large; long-lived; low reproductive rates; habitat specialists; living in open (wetlands) rather than closed (forest) habitats; rare; populations concentrated in few areas.

Visual impacts (s28–29)	Artificial lights / vehicle movement. Lighting can deter deer and provide insect food for bats but disrupt plant growth regulation, breeding/behaviour in birds, bats, nocturnal frogs, moths. **White (mercury vapour) lamps** attract insects → aerial-hawking bats → more collision mortality. Heavy trucks & high-speed trains: intensive discontinuous noise, vibration, visual disturbance that frightens mammals & birds.
Biogeochemical cycle (s30–31)	Nutrient/material flow determines ecosystem structure. Road development disrupts via alteration of surface & groundwater flows, **removal of biomass, relocation of topsoil**. Human activity can be a nutrient source → may increase **turbidity and BOD**. Impacts only roughly estimated based on: nature of alteration; data on soil erodibility; soil fertility; anticipated human activity; understanding of nutrient regime & energy flow.

Indirect impacts L8·s32–35 (World Bank 1997)

- **Accessibility:** roads increase human–nature contact → ecosystem modification (recreation, forest & mineral exploitation, colonization, urbanization); upgrading existing roads increases people having access.
- **Ecological disequilibrium:** importation of new plant/animal species along right-of-way upsets balance; native species face competition; non-native species gain advantage (lack of natural controls) → simplified, more vulnerable ecosystem.
- **Contamination of biota:** vehicles contaminate soil, air, water next to road (surface water well beyond); chronic contamination serious for top of food chain due to **bioaccumulation**.
- **Forest fires:** more human activity → more frequent fires; sudden, severe, wide-ranging (e.g. Dehradun, Uttarakhand; fires due to road accidents).

Crop yield L8·s36–37 (Burney & Ramanathan 2013)

- Rising temperatures and long-lived GHGs → significant negative impact on yields. Short-lived **black carbon and ozone** affect temperature, precipitation, radiation and can be toxic to plants.
- **Wheat yield in India in 2010 was 36% lower** than 1980s; some densely populated states had **50%** relative yield losses. **90%** of wheat-yield reduction attributed to **short-lived climate pollutants**. (Relative yield change 2006–2010 average for rice and wheat.)

Life-cycle stages of road/railway L8·s38 (TRANSGREEN 2019)

Stage	Potential problems
a Planning	Quality of routing
b Construction	Availability of biodiversity/ecological data; destruction of natural habitats; creation of new artificial habitats; effect on groundwater and surface water; mortality of animals at construction sites; noise, other emissions and contamination
c Operation	Barrier effect; mortality of animals, human casualties and damages; gaseous, liquid, solid emissions; noise pollution; road maintenance
d Removal	Processing and elimination of waste materials

Mitigation measures L8·s39–48 (Grilo et al. 2010)

Measure	Detail on slide
Fencing	Culvert with small-mesh fencing; underpass with **1.5 m fencing** to prevent animal entry onto roads.
Small underpasses	Metal or concrete pipe culverts for **smaller animals**
Intermediate underpasses	For **medium-sized animals** (foxes, black bear etc.)
Larger underpasses	For **large carnivores** and other animals
Extensive bridges	For large carnivores like **grizzly bears and wolves**
Wildlife overpasses	Allow **full sunlight and precipitation**; full vegetation growth similar to forests
Dry ledges	Box culvert with dry ledges on sides for smaller animals (stone marten, southern Portugal)
Earthen ramps (escape)	Escape from right-of-way accidents and for visibility of drivers
Amphibian tunnel	Top-graded; open top allows light and air flow
Paynes Prairie wall	Prevents turtles from accessing roads (Portland [as written])

- **Conclusion (s49):** transport can significantly harm flora & fauna; **careful planning is the only solution**.

Important Distinctions to Remember

- **Direct (8)** vs **Indirect (4)** impacts. Barrier effect = habitat fragmentation = corridor restriction concept family; four main effects: habitat loss, traffic mortality, population subdivision, barrier effect.
- Underpass size ↔ animal size: pipe culverts (small) → intermediate (medium: foxes, black bear) → larger (large carnivores) → extensive bridges (grizzly, wolves).
- Noise-sensitive: birds; avoiders: reindeer, elk, caribou, brown bear. Light: attracts insects/bats (white mercury vapour), deters deer.

Image-only slides of this lecture were rendered and read; their content is in **Part 2A (end of Part 2) under Lecture 8**.

Lecture 9 — Impacts of Transportation Systems-IV: Soil, Water, Acid Rain (53 slides)

Soil and water — physical impacts L9·s3–5

- Construction clears, levels, fills, cuts natural material → changes **soil density, landscape relief, surface and groundwater flows, microclimate** → alters land cover, vegetation, habitat composition.
- Wetlands & habitats especially sensitive to hydrology change (embankments, cuttings may drain aquifers, ↑ soil erosion, earth slides); can pollute watercourses with sediments; canalisation of surface water into ditches changes run-off and debris flows.

Chemical impacts L9·s6–8

- Pollutants: **road dust, salt, heavy metals, fertiliser nutrients, toxins**. Accumulate near infrastructure but effects on vegetation/fauna seen at distances of **several hundreds of metres**.
- **De-icing salts** (NaCl, $CaCl_2$, KCl, $MgCl_2$): damage vegetation, contaminate drinking water, **reduce soil pH** → **increases mobility of heavy metals**.
- **Heavy/trace metals** (Pb, Zn, Cu, Cr, Cd, Al from petrol, de-icing salts, dust) accumulate in plant/animal tissues; affect reproduction & survival.
- Exhaust toxins: **PAH, dioxins**, ozone, nitrogen, CO_2, fertilising chemicals. Emissions in lakes & heathlands at **>200 m** from road can change plant growth and species diversity.

Water quality — roads L9·s9–11

- Road drainage & excavation **lower** the water table; embankments & structures **raise** it by restricting flow. Can contribute to flooding, soil erosion, channel modification, siltation of streams. Road construction blocks subsurface streams → **wetting upflow side and drying downflow side**. Improper/inadequate drainage → rising water tables or flooding.

Water transport impacts L9·s12–20

- Marine transport emissions = **most important segment** of water-quality impact of transportation sector. Main effects from: **dredging, waste, ballast waters, oil spills**.
- **Dredging:** deepening harbor channels by removing sediments from the bed of a water body. Waste has very high bacteria level – hazardous to public health and marine ecosystems.
- Vessel emissions: air pollutants; water emissions – cargo hold cleaning water, ballast water, grey water, sewage, oil spills.
- **Ballast water** discharge from other parts of the world introduces **invasive species** → domestic species may go extinct (pollution different from emission type).
- **Oil spills:** mostly ship pollution; **less frequent than daily-operation pollution but devastating**; toxic to marine life. **PAHs** (components of crude oil) hard to clean, last years in sediment; cause developmental problems, disease susceptibility, abnormal reproductive cycles.

Spill	Facts
MV Wakashio (s17)	Bulk carrier aground off SE Mauritius, **August 7, 2020** (photo).
Haven 1991, Italy (s18)	Very large crude carrier HAVEN, **144,000 t** Iranian heavy crude; exploded due to **electric spark during tank cleaning**; >**10,000 t** spilled.
Gulsar Ana 2009, Madagascar (s19)	23,602 GT bulk carrier ran aground; **39,250 MT cargo, 568 MT heavy fuel oil, 66 MT diesel, 8 MT lube oils** lost; shore clean-up **120 days, 7500 man-days**.
RENA 2011, New Zealand (s20)	37,209 GT containership aground on **Astrolabe Reef**; **1368 containers** (11 dangerous goods); **1700 t IFO 380** bunker fuel + **61 t** diesel; ~**200 t** fuel oil released; >**1000** oiled birds, **250** live oiled birds incl. endangered penguins.

Air transport — water quality L9·s21–22

- Airports: jet fuel, lubricants, chemicals → water pollution. Cold climates: **de-icing fluids** fall to ground → storm-water runoff to streams/rivers/coastal waters if not recovered. Active ingredient: **ethylene glycol or propylene glycol** → high **BOD** during degradation → consume oxygen needed by aquatic organisms.

Acid rain L9·s23–29

- Formed when acidic pollutants **SO_2 and NO_2** dissolve in rain water.
- $SO_2(g) + H_2O(l) \rightleftharpoons$ **H_2SO_3** (sulphurous acid) → with O_2 slowly oxidized to **H_2SO_4** (sulphuric acid): $H_2SO_3 + O_2 \rightarrow H_2SO_4$ [as written]. NO_2 (with O_2, H_2O): $4NO_2 + 2H_2O \rightarrow 4HNO_3$ [as written] (**nitric acid**).
- **Pathway:** 1. SO_2/NOx emitted → 2. transformed into acid particles, transported long distances → 3. fall as **wet and dry deposition** (dust, rain, snow) → 4. harm soil, forests, streams, lakes.

Effect on	Detail
Land/soil	Soil acid levels rise → kills plants.
Plants	Stops trees and plants receiving oxygen [as written] → stops growth.
Material assets	Dissolves limestone and marble; erodes/destroys sculptures; buildings wither.
Water bodies	Lakes acidified easily due to **low water flow**; most aquatic species cannot survive → biodiversity affected.
Humans	**Only indirect effects.** SO_2 particles → chronic lung problems (asthma, bronchitis); NOx → ground-level ozone → severe lung problems (chronic pneumonia, emphysema).

Mitigation L9·s31–42 (World Bank 2019)

- Good road drainage features: cutoff/catch-water drain, scour checks, culvert, natural water course, side drain, turn-out drains (s31). Storm water inlets & sewers remove water from road surfaces (s32); highway surface drainage on sides and middle (s33).
- **Infiltration:** cross and side drains → groundwater recharge structures (s34); **roadside infiltration trenches with bunds** intercept surface runoff (s35); **tube recharge wells = Bhungroo in Gujarat, India** (s36); road water harvesting to surface storage (s38).
- **Irrigation from road drainage (s37):** runoff directed to interconnected soaking pits; upper bund breached then lower bund; repeat until all fields watered; smooth water movement to avoid erosion; fields well levelled. Examples: rice drainage via roadside borrow pit – Bangladesh (s39); grapes with road water – Bolivia (s40); collecting road water – Shandong, China (s41); urban lake "Ijzeren Vrouw" from borrow pit – Netherlands (s42).

Oil spill clean-up L9·s43–50

Method	Facts
Oil booms (containment)	Temporary floating barriers; reduce spread to shorelines, rivers, oceans
Oil skimmers	Recover spilled oil from water surface; trap floating oil
Sorbents	Absorb oil from water surface; clay, vermiculites or synthetic (plastic, fibres)
In-situ burning	(a) oil held by fire-resistant boom in segregated area; (b) ignition flame applied; (c) flame spreads, burns oil down to **1 mm thickness**; (d) fire ends when vapour insufficient to maintain combustion
Dispersants	Contain a form of **detergents**; break oil into smaller droplets, diluted in ocean
Manual labour	Time-taking; can take **months**
Hot water & high-pressure washing	Shorelines; removes stranded oil from hard surfaces (rocks, sea walls); effective but **can harm plants and animals** in treated zone
Bioremediation	Microorganisms use hydrocarbons (HC) as carbon source for metabolism; HC oxidized to **CO_2 and H_2O**

- **Conclusion (s51):** marine & aviation industries can also damage environment like road; **low-sulphur and alternate fuels** significantly reduce SOx and NOx (acid rain).

Important Distinctions to Remember

- Road drainage **lowers** water table vs embankments/structures **raise** it.
- Propylene/ethylene glycol (airport de-icing; BOD) ≠ road de-icing salts (NaCl etc.; reduce soil pH, mobilise heavy metals).
- $SO_2 \rightarrow H_2SO_3 \rightarrow H_2SO_4$; $NO_2 \rightarrow HNO_3$. Humans: only *indirect* acid-rain effects.

- Booms (contain) vs skimmers (recover) vs sorbents (absorb) vs dispersants (break into droplets) vs burning vs bioremediation.
- Haven 1991 (explosion, spark in tank cleaning) vs Gulsar Ana 2009 vs RENA 2011 vs Olivia 2011 (L8).

Image-only slides of this lecture were rendered and read; their content is in **Part 2A (end of Part 2) under Lecture 9**.

Lecture 10 — Impact of Transportation Systems-V: Noise (51 slides)

Noise and its effects L10·s3–8

- **Noise = unwanted sound**; level of sound exceeding acceptable level and creating annoyance. Main contributors: **urbanization and traffic**. Noise from vehicles ≈ **70%** of total traffic noise. Sources: road & rail traffic, aircraft, industry, construction.
- Effects in **three groups**: **Subjective, Behavioral, Physiological**.

Group	Description
Subjective	Annoyance, disturbance, noisiness; difficult to measure precisely (people perceive differently, e.g. music vs noise).
Behavioral	Interference with **speech, sleep** or other tasks; sleep disturbance in dense traffic areas, near railway tracks or airports.
Physiological	Sudden shock/frightening effect; headache; extremely high noise for long duration → **deafness**.

- **Harmful effects on humans (s8):** interferes with speech; emotional & behavioral stress; permanent hearing loss; diseases (headache, blood pressure, heart failure); health effects (increased heart beat, constriction of blood vessels, dilation of pupil); problematic for patients needing rest; potential to damage liver, brain and heart.

Noise level standards dB(A) L10·s9

Area	FHWA	AASHTO	CPCB day (6 AM–9 PM)	CPCB night (9 PM–6 AM)
1 Sensitive (parks, schools, hospitals, silent zones)	60	55–60	50	40
2 Residential	70 (interior max. 55)	70 exterior, 55 interior	55	45
3 Mixed	70	70	---	---
4 Commercial	75	75	65	55
5 Industrial	75	75	75	70

FHWA = Federal Highway Administration (USA); AASHTO = American Association of State Highway and Transportation Officials (USA); CPCB = Central Pollution Control Board (India).

Factors & sources of vehicular noise L10·s11–20

- Factors affecting traffic noise are grouped: **Traffic factors, Road factors, Vehicle factors, Human factors** (s11–16; contents are images). Human factors: irresponsible/inappropriate behavior; **90% of accidents** linked to human error (s17).
- **Facts (s13):** motor vehicles noisier than passenger cars [as written]; **diesel engines noisier than petrol**; commercial trucks & loaded vehicles noisier than other road modes; noise **directly proportional to age** of vehicle (older → more noise).
- **Two major vehicle sources (s18): Power plant & transmission** noise (engine, exhaust, intake, cooling system); **Running gear** noise (tyre–road interaction, differential, propeller shaft). On highways: at **lower speeds engine-exhaust noise dominant**; at **higher speeds road–tyre interaction dominant**.

Noise measurement L10·s21–30

- **Sound level meter block chain (s21):** microphone → transducer signal → **pre-amplifier** / filters → **weighing network** → **amplifier** → **rectifier** → **averaging** system → **display** (or AC outlet).
- **Acoustic arrays (s22):** number of microphones in various patterns; **omnidirectional**; receive sound from wide area.
- **Decibel (s23):** SPL (L_p) = logarithmic measure of effective pressure relative to reference; unit dB. $L_p = \ln(p/p_0)$ Np = $2 \log_{10}(p/p_0)$ B = **$20 \log_{10}(p/p_0)$ dB**; p = rms sound pressure; p_0 = reference. **1 B = 10 dB**. Reference in air **p_0 = 20 µPa** = threshold of human hearing (≈ mosquito flying 3 m away).
- **Leq (equivalent continuous noise level) (s24–26):** formula (s26): $L_{eq} = 10 \log[(1/(t_2-t_1)) \int (p_A^2/p_0^2) dt]$ from t_1 to t_2 (p_A = acquired sound pressure; p_0 = 20 µPa). preferred method for time-varying sound; single dB value

accounting for total sound energy over time. Example: over 24 h as low as **30 dB(A)** at night, **70 dB(A)** with occasional vehicles, higher by day. Meter stores fluctuations, calculates "average energy" – **not arithmetic average**: dB converted to SPL, summed, divided by number of samples, converted back to dB.

- **Energy average example (s27):** levels 88, 93 [text] and 97 dBA: arithmetic mean = (88+94+97)/3 = **93 dBA**; energy-average L_{eq} = 10 $\log_{10}$$\{[10^{8.8}+10^{9.4}+10^{9.7}]/3\}$ = **94.3 dBA**. *[Slide text writes "93" and "94" inconsistently; kept as on slide.]*
- **A-weighting dB(A) (s28–29):** most common weighting; cuts off lower and higher frequencies average person cannot hear; imitates human ear; ear responds more to **500 Hz–8 kHz**, less sensitive to very low/high pitch. Units dBA or dB(A).
- **n-percent exceeded level L_n (s30):** sound level exceeded n% of time. **L10 > L50 > L90. L10:** exceeded 10% of time – sporadic/intermittent events. **L50:** 50% – median / fluctuating noise. **L90:** 90% – **background/ambient** noise.

Mitigating measures L10·s31–33

- Green plantations (bushes/trees around sources) · Regular servicing & tuning of automobiles · Green buildings (noise-absorbing materials for walls, windows, ceilings) · Wearables (ear plugs, earmuffs) · Lubrication & regular servicing of machinery · Policy regulations (restrict loudspeakers in crowded/public areas) · Soundproof doors/windows · Land use (factories far from residential areas) · Community development programs (long-term planning) · Social awareness programs.

Noise barriers L10·s34–46

- Barriers reduce sound by blocking straight-line path source→receiver; reduction ≈ **5 dB**; further **3 dB** by making the barrier much higher (longer path) → combined **8 dB** = equivalent to reducing traffic by **factor of 6**. Diffraction depends on ratio of barrier height h to wavelength λ; **higher barrier → more efficient**.
- **Heights (s36):** min permissible **1–2.5 m**; max **4–10 m** or more; average (most common) **2–5 m**. Shadow zone behind barrier.

Barrier type	Slide comment
Timber acoustic fencing	Good quality stays imperforate over time
Hedging, foliage	Dense deep foliage gives some benefit; **line hedges ineffective**
Brick walls	Good, but expensive vs timber
Green screens	Gaps make them acoustically transparent
Transparent acoustic screens (glass/acrylic)	Good where visual amenity important
Hit & miss fencing; Trellising; Slatted fencing	Ineffective due to gaps
Concrete	Low maintenance but expensive
Light-weight metal fencing	Ensure minimum mass of **10 kg/m²**
Bunding	Suited to larger projects where spoil can be used
Gabion walls	Wall must be thick and aggregate small
Loosely lapped fencing	Some screening, gaps affect performance; gaps increase over time
Fully bespoke/engineered solutions	Larger infrastructure projects; very expensive

- **Transmission loss:** sound level reduced when forced to travel **through** the barrier; **30 dB** transmission loss = practically all (**99.9%**) sound blocked.
- **Insertion loss:** difference in sound level without barrier vs after barrier installed between source and receiver.
- **Insertion loss (IL), Kurze & Anderson model (s46):** IL = 5 dB + 20 log[√(2πN) / tanh√(2πN)] dB for N up to 12.5; **IL = 20 dB for N > 12.5**. **N = Fresnel number** = non-dimensional measure of how much farther the sound must travel to reach the receiver due to the barrier; **N = (a + b − l)·f / C_0**; l = original direct path length source→receiver; a, b = the two straight segments of the path modified by the barrier; f = sound frequency (Hz); C_0 = speed of sound in air (approx. **1125 ft/sec or 343 m/s**).
- **Numerical problem (s47–48):** 12 ft barrier, 12 ft from vehicle; house 15 ft beyond barrier; window height 4 ft. l = √(27^2+4^2) = **27.3 ft**; a = √(12^2+12^2) = **17 ft**; b = √(15^2+8^2) = **17 ft**; a+b−l = 34 − 27.3 = **6.7 ft**. At f = 100 Hz, **N = (6.7 × 100)/1100 = 0.61** [slide uses 1100]; IL = 5 + 20 log[√(2π×0.61)/tanh√(2π×0.61)] ≈ **10 dB**.

- **Conclusion (s49):** loud, regular exposure → effects from slight disturbance to even death; reduce noise from planning stage; trees/shrubs and barriers help.

Important Distinctions to Remember

- **Subjective** (annoyance) vs **Behavioral** (speech/sleep interference) vs **Physiological** (shock, headache, deafness).
- **Leq** = energy average, not arithmetic mean (94.3 vs 93 dBA). L10 (intermittent) > L50 (median) > L90 (background).
- **Transmission loss** (through barrier; 30 dB = 99.9% blocked) vs **insertion loss** (with vs without barrier).
- Low speeds: engine-exhaust noise dominates; high speeds: tyre–road dominates. Diesel > petrol; older vehicle = noisier.
- Barrier: 5 dB + 3 dB = 8 dB ≈ traffic ÷ 6. Reference pressure 20 μPa; 1 B = 10 dB.

Image-only slides were rendered and read (Part 2A, Lecture 10) **except s11, s12, s14, s15, s16**: the factor lists for traffic noise are not present in the PDF (headings and pictures only).

Part 2A — Content recovered from image-only slides

These slides had no text layer. I rendered each one and read it. Values are only those visible on the slide. Charts are given as stated labels/values; where a chart has no printed values, only its visible trend or labels are noted. Tags: L·s = lecture · slide. Slides not recovered are listed at the end.

Lecture 1

- L1·s47 **Timeline "Evolution of Transport Technology since the 18th Century"** (Rodrigue 2020). Four rows with milestones (axis: 1800, 1900, 1950, 2000; labels under axis: **Steam engine** (1800 area), **ICE** (1900 area), **Automation** (2000)).
  Air: Balloons · Airships · Planes · Helicopters · Jet planes · Jet engine · Wide-body jets · Supersonic jet · Drones.
  Rail: Steam · Locomotives · Subways · Tramways · High speed rail · Maglev.
  Road: Omnibus · Bicycles · Surfaced roads · Automobiles · Trucks · Buses · Highways · Electric cars · Hybrid cars · Self-driving vehicles.
  Maritime: Locks · Docks · Steamships · Iron hulls · Liners · Bulk ships · Supertankers · Containerships.
  [Slide is displayed mirrored/upside-down in the PDF; exact horizontal positions of each item not reliably readable.]

Lecture 3

- L3·s16 **Interaction of Economy on Ecological System:** SUN → solar energy flows into the **Global Ecological System**, which contains **Non-Renewable Resources** and **Renewable Resources** feeding the **Economic subsystem**; economic subsystem → **Waste** → **Recycling** (back to economy) and **Damages**; **Waste Heat** leaves; time axis shown.
- L3·s19 **17 SDGs:** 1 No Poverty · 2 Zero Hunger · 3 Good Health and Well-being · 4 Quality Education · 5 Gender Equality · 6 Clean Water and Sanitation · 7 Affordable and Clean Energy · 8 Decent Work and Economic Growth · 9 Industry, Innovation and Infrastructure · 10 Reduced Inequalities · 11 Sustainable Cities and Communities · 12 Responsible Consumption and Production · 13 Climate Action · 14 Life Below Water · 15 Life on Land · 16 Peace, Justice and Strong Institutions · 17 Partnerships for the Goals.
- L3·s22 **Sustainable Transport Goals (Venn, UNDESA 2014): Economic:** efficient mobility; local economic development; operational efficiency. **Social:** social equity (fairness); human safety and health; affordability; community cohesion; cultural preservation. **Environmental:** air, noise and water pollution reductions; climate change emissions; resource conservation; open-space preservation; biodiversity protection. (Icons: electric car, carpooling, bike lane, train.)
- L3·s26 **"5 Is and 5 Cs" approach (UNDESA 2014):**

5 I's — areas of action	5 C's — to meet the needs of
Infrastructure – enabling services	**Client** – meeting expectations with quality services
Innovation – for the future	**Centres** – passenger and logistic hubs
Integration of modes	**Corridors** – connecting the hubs
Intelligence – capacity building & training	**Congestion** – low pollution improved efficiency
Investment – to create a virtuous cycle	**Complementarity** – optimized multimodal transport systems

- L3·s27–29 **Goals, Objectives & Performance Indicators (Todd Litman 2021 / vtpi):**

Goal	Objectives	Performance indicators
I. Economic (s27)		
Economic productivity	Transport system efficiency; transport system integration; maximize accessibility; efficient pricing and incentives	Per capita GDP; portion of budgets devoted to transport; per capita congestion delay; efficient pricing (road, parking, insurance, fuel etc.); efficient prioritization of facilities
Economic development	Economic and business development	Access to education and employment opportunities; support for local industries
Energy efficiency	Minimize energy costs, particularly petroleum imports	Per capita transport energy consumption; per capita use of imported fuels

Affordability	All residents can afford access to basic (essential) services and activities	Availability and quality of affordable modes (walking, cycling, ridesharing, public transport); portion of low-income households that spend **more than 20%** of budgets on transport
Efficient transport operations	Efficient operations and asset management maximizes cost efficiency	Performance audit results; service delivery unit costs compared with peers; service quality
II. Social (s28)		
Equity / fairness	Transport system accommodates all users incl. those with disabilities, low incomes, other constraints	Transport system diversity; portion of destinations accessible by people with disabilities and low incomes
Safety, security and health	Minimize risk of crashes and assaults; support physical fitness	Per capita traffic casualty (injury and death) rates; traveler assault (crime) rates; human exposure to harmful pollutants; portion of travel by walking and cycling
Community development	Helps create inclusive and attractive communities	Land use mix; walkability and bikability; quality of road and street environments
Cultural heritage preservation	Respect and protect cultural heritage; support cultural activities	Preservation of cultural resources and traditions; responsiveness to traditional communities
III. Environmental (s29)		
Climate stability	Reduce global warming emissions; mitigate climate change impacts	Per capita emissions of GHGs (CO_2, CFCs, CH_4 etc.)
Prevent air pollution	Reduce air pollution emissions; reduce harmful pollutant exposure	Per capita emissions (PM, VOCs, NOx, CO etc.); air quality standards and management plans
Minimize noise	Minimize traffic noise exposure	Traffic noise levels
Protect water quality & hydrologic functions	Minimize water pollution; minimize impervious surface area	Per capita fuel consumption; management of used oil, leaks and stormwater; per capita impervious surface area
Openspace and biodiversity protection	Minimize transport facility land use; encourage compact development; preserve high quality habitat	Per capita land devoted to transport facilities; support for smart growth development; policies to protect high value farmlands and habitat
IV. Good Governance and Planning (s29)		
Integrated, comprehensive and inclusive planning	Clearly defined planning process; integrated and comprehensive analysis; strong citizen engagement; lease-cost [as written] planning	Clearly defined goals, objectives and indicators; availability of planning information and documents; portion of population engaged in planning decisions; range of objectives, impacts and options considered; efficient and equitable funding allocation

Lecture 4

- L4·s5 Registered vehicles across India (millions): 2001 = **55**; 2002 58.9; 2003 67; 2004 72.7; 2005 81.5; 2006 89.6; 2007 96.7; 2008 105.4; 2009 115; 2010 127.7; 2011 141.9; 2012 159.5; 2013 176; 2014 190.7; 2015 210; 2016 230; 2017 253.3; 2018 272.6; **2019 = 295.8**. Yearly growth rate (%): 2002 7.1; 2003 13.7; 2004 8.5; 2005 12.1; 2006 9.9; 2007 7.9; 2008 9; 2009 9.1; 2010 11; 2011 11.12; 2012 12.4; 2013 10.34; 2014 8.35; 2015 10.12; 2016 9.52; 2017 10.13; 2018 7.62; 2019 8.51.
- L4·s6 Stacked bar (2011) for **Delhi, Mumbai, Hyderabad, Bangalore**; categories: goods vehicles, buses, taxis, auto, two-wheelers, cars, jeeps, omni buses, others. Two-wheelers are the largest share (~60–70% in the bars), cars next.
- L4·s8 Composition of registered vehicles (% of total), 1951→2016: **Two wheelers** 8.8 (1951) → 13.2 → 30.9 → 48.6 → 66.4 → 70.1 (2001) → 71.8, 72.4, 72.7, 73.1, 73.5, 73.5 (2016). **Cars, jeeps & taxis** 52 (1951) → 46.6 → 36.6 → 21.5 → 13.8 → 12.8 (2001) → 13.6 … 13.1 (2016). **Buses** 11.1 (1951) → 8.6 → 5 → 3 → 1.5 → 1.2 … 0.8 (2016). **Goods vehicles** 26.8 (1951) → 25.3 → 18.4 → 10.3 → 6.3 → 5.4 … 4.6 (2016). **Other** 1.3 (1951) → 6.3 → 9.1 → 16.6 → 11.9 → 10.5 … 8.1 (2016). *[Some intermediate points read from chart labels.]*
- L4·s9 Registered motor vehicles in Delhi (millions): **1.42 (1988)**, 1.76 (1990), 2.37 (1994), 2.79 (1996), 3.17 (1998), 3.46 (2000), 3.93 (2002), 4.58 (2004), 5.14 (2006), 5.9/6.01 (2008/09), **6.93 (2011)**, 7.79 (2013), 8.83 (2015), 9.7 (2016), 10.99 (2018), **11.89 (2020)**.

- L4-s17 GHG from transport 2018 (million metric tons CO_2e): **United States 1,762.23; China 917.02; India 305.33 (3rd)**; Russia 258.86; Japan 204.56; Canada 192.11; Brazil 191.66; Germany 158.3; Mexico 156.58; Indonesia 154.01; Iran 139.23; Saudi Arabia 136.02; France 125.42; United Kingdom 120.66.
- L4-s18 Transport emission per capita, G20, 2018 (excl. aviation) and % change 2013–2018: **TUR +38%**, **IND +28%**, **CHN +23%**, KOR +14%, EU +7%, G20 +6%, GER +4%, USA +4%, AUS +2%, UK +2%, IDN +2%, FRA −2%, MEX −3%, JPN −5%, ITA −5%, CAN −5%, RUS −6%, ARG −6%, ZAF −6%, BRA −11%, **SAU −25%**; OECD +4%; Non-OECD +9%. India's per-capita bar is among the lowest (marked on slide). USA and Australia/Canada have highest bars (~4–5.3 tCO_2/capita). *[Slide 19 text says "India and China 2nd and 3rd largest contributors" — consistent with +28% and +23%.]*
- L4-s24 Smart city illustration (Esri India) with alert labels: Multi-modal authentication; Air pollution index; Petrol fumes alert; Graffiti/fence breached; Crowd detected; Vehicle of interest alert; Person of interest alert; Mega event integrated security.
- L4-s31 Modal share of NMT (walking + bicycle) by city (1980, 1995, 2000, after 2005) — bar chart without printed values; **Chennai and Patna** rise/hold while others (e.g. Delhi, Bangalore, Ahmedabad, Kanpur, Jaipur) fall.
- L4-s32 Bicycle trips chart (Delhi, Chennai, Ahmedabad, Kanpur, Jaipur, Patna; 1980, 1990, 2000, after 2005): arrows show **Delhi −58%**, **Jaipur −86%**, **Patna +83%** — matches the bullet text.
- L4-s36 **Fund allocation:** Others **45%**, Roads **44%**, Mass transit **11%**. **Consequences of investments in roads (cycle):** More roads → Increased private modes → Traffic congestion → (back to) More roads.
- L4-s38 Cost and capacity of mass transit systems: x-axis capacity (passengers/hour/direction, 0–90,000), y-axis capital cost per km (M USD). Qualitatively: **MRT** highest capacity and highest cost (~up to 120 M USD/km, capacity ~60,000 central); **Monorail** ~60 central (range ~35–100); **LRT** ~35–40 M USD/km, capacity ~20,000; **BRT/busway & high-capacity BRT** lowest cost (~10–20 M USD/km).
- L4-s39 Modal share pre/post COVID-19 (TERI survey, April 2020; % of respondents): **Metro** falls (≈25% → ≈17%); **Personal car** rises (≈24% → ≈30%); Autorickshaw ≈ unchanged (~11%); Two-wheeler ≈10% → slightly less; **Bus** falls (≈10% → ≈6%); Taxi up; Cycle up; Walking up; Shared cab ≈ flat/slightly down; Carpooling up; Local train down; Company vehicle ≈ up slightly; No travel/work from home up; Shared bike very small. *[Approximate bar readings; no printed values.]*
- L4-s40 Distance to work (0–5, 5–10, 10–20, >20 km) vs post-COVID mode: **Non-motorized transport** users are those with shortest distances (about two-thirds 0–5 km, rest 5–10 km); **Still use public transport**, **Company vehicle/shared cab**, **Private vehicles** mostly >20 km; Autorickshaw = 5–10 and >20 km only.
- L4-s41 Age (18–30, 30–45, 45–60, above 60) vs post-COVID mode: **Carpooling** = 100% aged 18–30; Still use PT mostly 18–30 (~75%); NMT is the most evenly spread across ages (18–30 ≈ one third, 30–45 ≈ one third, 45–60 ≈ one third).

Lecture 5

- L5-s7 Average global temperature (≈13.6–14.6 °C axis) and carbon emissions from fossil fuel burning (million tons of carbon, ~0 to ~10,000) 1800–2014: both rise, steepest after ~1950.
- L5-s10 GHG table verified (see Lecture 5 table): **CO_2 lifetime "Varies"; GWP 1 · CH_4 12.4 years, GWP 28–36 · N_2O 121 years, GWP 265–298 · Fluorinated gases: few weeks to 1000 years, GWP "varies upon the gas"**.
- L5-s11 Concentration charts 1950–2015: CO_2 (ppm, rising to ~400+), CH_4 (ppb, ~1,100 → ~1,800), N_2O (ppb, ~285 → ~325), ozone (Dobson units; total column roughly flat ~275–290; stratosphere vs troposphere shown).
- L5-s13 Radiative forcing by human activities (W/m^2): long-lived GHGs – **CO_2 (largest warming, ~1.6)**, **methane** (~1), halogenated gases (slightly negative: "negative radiative forcing is because some of these gases destroy ozone"), nitrous oxide; short-lived gases that create ozone etc.; **aerosols** (net cooling; **black carbon (soot)** = warming); changes in clouds due to aerosols (cooling); change in albedo due to land use (albedo = reflectiveness of Earth's surface); natural: change in energy from the sun (very small); **net total due to human activities ≈ +2.3 (warming)**.
- L5-s23 Arctic sea ice: NASA images, **September 1979 vs September 2015** — dwindling ice.
- L5-s24 Arctic sea ice extent 1979–2016 (million sq miles): **March (max) ≈ 6.4 → ≈5.6**; **September (min) ≈ 2.8 → ≈1.8** (low ≈1.4 around 2012). Age of ice at minimum September week 1983–2015: first-year, second-year, third-year, fourth-year, 5+ year ice; older ice declines; first-year ice share rises.
- L5-s36 **Summary of potential impacts of climate change** (Cambridge Books Online 2010): drivers **Temperature, Sea-level rise, Precipitation** → **Health** (weather-related mortality, infectious diseases, air-quality respiratory illnesses); **Agriculture** (crop yields, irrigation demands); **Forest** (forest composition, geographic

range of forest, forest health and productivity); **Water resources** (water supply, water quality, competition for water); **Coastal areas** (beach erosion, inundation of coastal lands, additional costs to protect coastal communities); **Species and natural areas** (loss of habitat and species; cryosphere: diminishing glaciers).

- L5·s37 **Number of disasters per type 1998–2017 (UNISDR):** Flood **3,148 (43.4%)**; Storm **2,049 (28.2%)**; Earthquake 563 (7.8%); Extreme temperature 405 (5.6%); Landslide 378 (5.2%); Drought 347 (4.8%); Wildfire 254 (3.5%); Volcanic activity 99 (1.4%); Mass movement (dry) 12 (0.2%).
- L5·s38 Disasters per year by category 1998–2017: **climate-related ≈ 290–440 per year** (peak ≈ 2005) vs **geophysical ≈ 25–50 per year**.
- L5·s39 Annual disaster deaths 1998–2017: geophysical spikes ≈ **2004 (~230,000)** and **2010 (~230,000)**; climate-related peak ≈ **2008 (~150,000)**, ~80,000 (2003), ~90,000 (2010). *[Slide's caption says climate-related deaths are "more in the last decade"; the chart's biggest spikes are geophysical.]*
- L5·s40 Map "Changes in CO_2 emissions caused by COVID-19 confinement measures" (1 January 2020 frame; scale −40% to 0%; Global Carbon Project / Le Quéré et al.) — "This improvement is only temporary!!!"

Lecture 6

- L6·s5 Spatial & durational effects of pollutants (duration vs space): **Lead, Noise** = local & short duration; **Particulates** = local; **NOx** = regional; **CO_2** = global & longest duration.
- L6·s7 Environmental dimensions: **Causes** (Land use ↔ Economics) → **Activities** (infrastructure construction and maintenance; vehicle and parts manufacture; travel; vehicle maintenance and support; disposal of vehicles and parts; other) → **Outputs** (habitat changes; emissions; ambient levels; exposure) → **End results** (social or ecological effects; health, environmental or welfare effects).
- L6·s9 Global energy systems transition (% of market, 1850–2150): Solids (wood, coal) decline; Liquids (oil) peak ≈ 2000 at ~45%; Gases (natural gas → hydrogen) rise toward ~100% by 2150.
- L6·s15 Average CO_2 emissions: **Passengers (g CO_2 per pass-km):** Large cars ≈ 195; Aviation ≈ 130; Average cars ≈ 130; Buses ≈ 45; **Rail ≈ 20 (lowest)**. **Freight (g CO_2 per ton-km): Aviation ≈ 435 (highest)**; Medium trucks ≈ 95; Heavy trucks ≈ 80; Rail ≈ 15; **Maritime ≈ 8 (lowest)**. *[Approximate bar readings.]*
- L6·s18 **Environmental externalities of transportation** (Rodrigue 2020): **Supra and infrastructure / Labor productivity / Agricultural productivity** ← vehicle emissions: CO_2 (30%), CO (70–90%), SO_2 (5%), NOx (45–50%), HC/VOC (40–50%), particulates (25%), O_3 (indirect), smog (indirect), acid rain (10–30%), CFCs (20%), lead (30%). **Commercial fishing / recreational facilities / water purification** ← air pollution fallouts; marine vessel discharges and spills; de-icing of infrastructure; runoffs; construction and maintenance; **dredging (80%)**. **Accidents/spills** ← marine vessel spills, accidental and intentional releases (hazmat): **Road (84%), Rail (12%), Air (3%), Maritime (1%)**. **Property values** ← vehicle emissions: **Road (70%), Rail (10%), Air (20%)**. **Public health** ← vehicle emissions during transport. **Damage to ecosystems** ← vehicle emissions, during transport, infrastructure.
- L6·s20 **Externalities of air pollution: Economic costs** – infrastructure (loss of useful life/amortization; replacement and restoration), labor productivity (people-hours-wage losses), agricultural productivity (yield decrease; biomass restoration time). **Social costs** – public health (medical services costs; burden of diseases). **Environmental costs** – damage to ecosystems (biological diversity and sustainability).
- L6·s24 **Impacts of climate change on transportation:**

Climate element	Operations	Infrastructures
Heat waves	Lift-off load limits on shorter runways; limits on periods of construction activity	Thermal expansion of bridges; pavement integrity and softening; deformation of rail tracks
Rising sea levels / intensity of precipitation	Frequent interruptions of coastal low-lying road, rail, air traffic due to storm surges; more weather-related delays and disruptions, particularly road and air	More frequent flooding of infrastructure in low-lying areas; erosion of infrastructure support; changes in harbor facilities for higher tides and surges
More frequent hurricanes	Frequent interruptions of air services; frequent and extensive evacuations of coastal areas; debris of road and rail infrastructures	Greater probability of infrastructure failure; greater damage to port infrastructures
Increase in Arctic temperatures	Longer shipping season; more ice-free ports in northern regions; availability of trans-arctic shipping routes	Damage because of thawing of the permafrost; shorter season for ice-roads

- L6·s25 Carbon cycle (GtC pools / fluxes): atmosphere (800), +3 [120+3 photosynthesis]; plant biomass (550); soil (2300); surface ocean (1000); deep ocean (37,000); fossil pool (10,000); reactive sediments (6000); fluxes: plant respiration 60, microbial respiration/decomposition 60, air–sea exchange 90 / 90+2, **fossil fuels, cement and land-use change 9** (red = human emissions), **atmospheric carbon net annual increase 4**, net terrestrial uptake 3, net ocean uptake 2.

Lecture 7

- L7·s6 **Transportation-induced health impacts (Khreis 2017):** Urban transport policy and planning → Built environment ↔ Transport provision and mode choice → outcomes list: Motor vehicle crashes · Air pollution · Noise · Heat islands · Lack of green space and biodiversity loss · Physical inactivity · Social exclusion · Community severance · Climate change → **Premature mortality** and **Morbidity**.
- L7·s7 **Planning decisions chain (Litman 2013):** Planning decision (infrastructure investment, parking requirements, fees and taxes, traffic management) → Land use patterns (development location, density and mix, parking supply and price, building orientation) → Travel behavior (amount of walking, cycling, public transit, vehicle ownership, automobile travel) → Economic, social and environmental impacts (consumer costs, public service costs, crashes, pollution emissions, physical fitness).
- L7·s14 **Vehicle emissions and efficiency (EEA 2016):** fossil-fuel road transport = most significant source of transport-related air pollution. Fuel energy 100% → **engine losses 70%** (thermal 60%, combustion, pumping, friction); **parasitic losses 5%**; **drivetrain losses 5%**; power to wheels **20%** (rolling resistance 5%, wind resistance 10%, braking 5%). Emission sources: refueling losses (HC, VOC); evaporating emissions (HC, VOC); exhaust (CO_2, CO, HC, NOx, PM); abrasion of tyres, brakes and clutch (PM); road suspension wear (PM); re-suspension of road dust (PM).
- L7·s30 Deaths due to air pollution, 2019 (IHME): **China 1,848,000; India 1,667,000**; Pakistan 235,700; Nigeria 197,600; Indonesia 186,300; Bangladesh 173,500; Egypt 91,700; Russia 77,500; Ethiopia 77,000; Philippines 74,800; Myanmar 74,500; Vietnam 71,700; Brazil 60,900; United States 60,200; Japan 42,600; Poland 31,100; Germany 29,300; Italy 27,900; UK 15,000; France 14,300; Spain 11,800.
- L7·s31 Leading causes of death worldwide 2019 (% , WHO 2021): Ischaemic heart disease 16; Stroke 11.2; COPD 5.8; **Lower respiratory infections 4.7**; Neonatal conditions 3.7; **Trachea, bronchus, lung cancers 3.2**; Alzheimer disease and other dementias 3; Diarrhoeal diseases 2.7; Diabetes mellitus 2.7; Kidney diseases 2.4.
- L7·s32 **DPSEEA framework (WHO): Driving force** – urbanization, migration, transport policy (action: economic, transport policies). **Pressure** – rise in number of vehicles; increase commuting by vehicles; emission of pollutants, noise (action: emission control). **State** – congested roads, hazardous driving situation; decrease physical activity; increase lead & toxic gases, noise (action: environmental improvement). **Exposure** – risk for road traffic accidents; sedentary lifestyle, less energy expenditure; exposure to pollutants, noise (action: awareness, education). **Effect** – road traffic injuries, road rage; obesity, diabetes, CVD; respiratory disease, noise-induced health problems (action: treatment).

Lecture 8

- L8·s4 Schematic of impacts of transport infrastructure on ecology (Seiler 2003): **Barrier, Traffic mortality, Pollution, Conduit, Corridor, Land uptake, Habitat transformation, Avoidance, Predation, Disturbance/Edge effects**.
- L8·s6 Wildlife population over time after road construction: low risk of extinction pre-road mortality; during **time lag**: habitat loss → reduced habitat quality → wildlife mortality → reduced connectivity; after: **high risk of extinction, post-road mortality**; population falls from ≈200 to ≈50/20.
- L8·s8 Road density illustration: **0.5, 1.0, 2.0, 3.0, 4.0 km/km²**; as density rises, undisturbed habitat (white) shrinks; "grey corridors" = disturbance effect zones.
- L8·s13 Barrier effect graph vs traffic intensity (adt, 0–10,000): **attempted crossings** fall from 100% to ≈40%; **successful crossings** fall to ≈0; **killed in traffic** rises then levels at ≈38%.
- L8·s16 Factors influencing number of wildlife traffic accidents: **positive (+):** traffic intensity, vehicle speed, animal abundance; **negative (−):** awareness of driver, mitigation measures; **mixed (+/−):** time of year/day, road(side) attractiveness, habitat juxtaposition, integration in landscape, roadside vegetation.

Lecture 9

- L9·s5 Photo: interrupted subsurface water flow due to road openings (World Bank 2019).

- L9·s30 **Sources of transportation air pollution (EPA): On-road:** light-duty vehicles, heavy-duty vehicles, motorcycles. **Nonroad:** aircraft, nonroad diesel equipment, marine engines/ocean vessels, locomotives, lawn & garden, recreation vehicles. Smog and soot → health and welfare impacts; CO_2 and other GHGs → climate change. **Solutions (emission reductions → cleaner air & better health):** catalytic converters (with unleaded gasoline and low sulfur levels reduce hydrocarbon & nitrogen oxide emissions); fuel standards (reduce exposure to pollutants like lead and benzene; renewable fuels reduce CO_2); engine technologies (computer controls, variable valve timing, multi-valve engines, turbo charging, gasoline direct injection); transmission technologies (7+ speeds, dual clutch transmissions, continuously variable transmissions); diesel filters (reduce particulate matter from on-road and off-road diesel engines); alternative vehicle technologies (plug-in electric vehicles & fuel cells = zero tailpipe emissions); better transportation planning (for passengers & freight reduces emissions & fuel use).
- L9·s46 In-situ burning steps confirmed: (a) oil collected/held by fire-resistant boom in segregated area; (b) ignition flame applied; (c) flame spreads, burns oil down to 1 mm thickness; (d) fire ends when vapour insufficient to maintain combustion.

Lecture 10

- L10·s9 **Noise level standards dB(A) (verified):**

Area	FHWA	AASHTO	CPCB day (6 AM–9 PM)	CPCB night (9 PM–6 AM)
1 Sensitive (parks, schools, hospitals, silent zones)	60	55–60	50	40
2 Residential	70 (interior max. 55)	70 exterior, 55 interior	55	45
3 Mixed	70	70	---	---
4 Commercial	75	75	65	55
5 Industrial	75	75	75	70

- L10·s10 **Typical noise levels dB(A):** 120 aircraft at take off; 110 car horn; 100 subway; 90 truck, motorcycle; 80 busy crossroads; 70 noise level near a motorway; 60 busy street through open windows; 50 light traffic; 30 quiet room; 10 desert; 0 hearing threshold. Bands: **Extremely loud** (above ~100), **Very loud** (~80–100), **Loud** (~60–80), **Moderate** (~40–60), **Faint** (below ~40).
- L10·s11, 12, 14, 15, 16 "Factors affecting Traffic Noise" (overall; Traffic, Road, Vehicle, Human factors): in the PDF these slides contain **only pictures and the headings** — the factor lists appear to have been text boxes not preserved in the PDF (rendered slides show blank areas). Pictures: traffic (cars, honking truck); road (smooth road, cobbled road, pothole, tall buildings, noise barrier wall); vehicle (many vehicle types, fuel pump icon, maintenance/spanner icon, no-horn sign); human (drunk-driving driver, reckless driver). **No factor list could be recovered — please check your original PPT.**
- L10·s19–20 **Rail noise sources:** aerodynamic noise; propulsion system; wheel–rail interaction. **Aircraft noise sources:** aircraft structure noise; wing; landing gear; engine ("noise caused by different types of engines").
- L10·s23 Decibel formula: $L_p = \ln(p/p_0)$ Np $= 2 \log_{10}(p/p_0)$ B = **$20 \log_{10}(p/p_0)$ dB** (as in notes).
- L10·s26 **Leq formula:** $L_{eq} = 10 \log [(1/(t_2-t_1)) \int_{t1}^{t2} (p_A^2/p_0^2) dt]$; L_{eq} = equivalent continuous sound pressure level in dB; p_0 = reference pressure (typically 20 µPa); p_A = acquired sound pressure; t_1 = start time; t_2 = end time. The equation divides the integrated, normalized sound pressure by the duration of interest; result in decibels.
- L10·s44 Noise barrier effectiveness diagram: Incident, Reflected, Direct, Diffracted paths; **Shadow zone** behind the barrier (labels only).
- L10·s46 **Insertion loss (Kurze & Anderson):** IL = 5 dB + 20 log [$\sqrt{(2\pi N)}$ / $\tanh\sqrt{(2\pi N)}$] dB for N up to 12.5; **IL = 20 dB for N > 12.5**. **$N = (a + b - l) f / C_0$** (Fresnel number).
- L10·s48 Worked example: a + b − l = 34 − 27.3 = 6.7 ft; at f = 100 Hz, **$N = (6.7 \times 100)/1100 = 0.61$** [slide uses 1100 for C_0]; IL = 5 + 20 log [$\sqrt{(2\pi \times 0.61)}$ / $\tanh\sqrt{(2\pi \times 0.61)}$] ≈ **10 dB**. Diagram: barrier 12 ft high; 12 ft from source to barrier; 15 ft barrier to house.

Still not recovered: L10 s11, s12, s14, s15, s16 (factor lists absent from the PDF itself). Photo-only slides (L1, L2, L8 s39–48, L9 s31–42 etc.) hold only captions already in the notes. Chart values marked "≈" are approximate readings of bars/lines — they

are not printed on the slide.

Part 3 — Rapid-Revision Sheet (all 10 lectures)

Memory aid only — the full facts are in Part 2.

Key dates / names (L1)

Year	Event
8000 BCE	Log rafts; Pesse Dugout Canoe
3500 BC	Wheel (Mesopotamia; pottery)
1490	Da Vinci flying machine plans
1769	Cugnot steam vehicle – first true automobile
1783 (Dec 1)	Charles & Robert – first manned gas-balloon ascent
1817	Draisienne (Karl Drais) – first two-wheeler
1819	SS Savannah – first steamship across Atlantic
1825	Stockton & Darlington Railway; Cheshunt monorail (horse)
1826	Omnibus
1830s–1930s	Great Railway Era
1853	Cayley glider – first successful manned flight
1860s	Velocipede / boneshaker (Michaux)
1876	Steam monorail (Philadelphia); electric tram (title 1876 / text 1879)
1883	Penny-farthing
1880s	Rover Safety (J. K. Starley); coal powers first electric generator (1880)
1885	Daimler Reitwagen motorcycle; Benz car (Daimler 1886, Hammel 1888)
1896	Pilcher Hawk
1902 / 1905	Wright glider / Wright Flyer
1908 / 1912	Model T / Titanic (Apr 10)
1919	Vickers Vimy – first nonstop transatlantic
1935	DC-3
1959–2000	Hovercraft (40 years)
1966	Bell 206-B (4 passengers)
1969 / 1976 / 2003	Concorde first flight / service / end; Tu-144 June 5 1969
1987 / 1988	A320 first flight / service (150 pax)
1999 / 2003	Global Express (11,100 km) / Maglev

Definitions & lists

- **Transportation system** = elements + interactions producing demand for travel and supply of transport services (Cascetta 2001). Multimodal = Combined/Intermodal transport; integration of vehicle mode, service, infrastructure.
- **11 efficiency factors:** Speed, Safety, Adequacy, Frequency, Regularity, Integration, Responsibilities, Comfort, Cheapness, Fuel efficiency, Employment generation.
- **Development factors:** Environmental, Historical, Technological, Political, Economic × Local, Regional, National/Transnational, Global.
- **Sussman components:** Government, Competition, Supply industry, Stakeholders, Financial community (+ customer, general public).
- **Sustainable development (Brundtland 1987)** – needs of present without compromising future generations. Pillars: Social, Environmental, Economic. Principles: environmentally bearable, socially equitable, economically viable. Stages: before 1972 / 1972–1987 / after 1987.
- **Sustainable transport (UNDESA 2014)** – safe, affordable, accessible, efficient, resilient + minimizing carbon/other emissions. SDGs: 2, 3, 7, 9, 11, 12, 13. "5 Is and 5 Cs".
- **GWP** converts gases to CO_2 equivalents; **radiative forcing** = heating effect of GHGs.
- **Impact types (L6):** direct / indirect (higher consequence) / cumulative.

- **Health (L7):** 3 broad ways – accidents, pollution, infectious diseases. Strategies: traffic calming, active transport, public transit, pricing reforms, mobility management marketing, smart growth. DPSEEA.
- **Flora/fauna (L8):** Direct 8 + Indirect 4; four main effects; mitigation crossing structures by animal size.
- **Noise (L10):** Subjective / Behavioral / Physiological; Leq, A-weighting, L10>L50>L90; transmission vs insertion loss.

Numbers to memorise

Lecture	Numbers
L3	Hong Kong 93% (PT 48 + walk 45), Octopus 1997 · DFC >450 Mt CO_2/30 yr · Johannesburg BRT US$749 m, returns US$900 m (2026 as written) · Copenhagen CO_2-neutral 2025, 96% <15 min, >50% cycle
L4	Urban 31.2% (377 m) 2011 → 40% (590 m) 2030 → 58% (875 m) 2050 · 63% GDP · vehicles +438% (2019 vs 2001), 10%/yr · 0.3 m (1951) → 230.03 m (2016) · vehicles +219% vs urban roads +124% · Mumbai 2nd, Bengaluru 6th, Delhi 8th, Pune 16th of 416 cities/56 countries · 151,000 deaths (414/day), 6% global, 70% young · potholes 4770 · 4.2 m global deaths, 1.1 m India · GHG: energy 68.7, agri 19.6, industry 6, LULUCF 3.8, waste 1.9 · road >90% transport GHG · PT 18.1% work trips · Smart Cities 100 / AMRUT 500, $8 bn/$8.3 bn · Delhi Metro 213 km, 2.8 m/day · 1.6 m buses; 170,000 public buses, 70 m/day · NURM 2009 · BRT 222+282 km; 504 km, 11 cities · hybrid bus INR 23 m vs diesel 9 m · NMT 40–60% (1980s) · Delhi metro 24% rickshaw feeder
L5	Sun 11-year cycle · 7 warmest since 2014, 10 since 2005 · SST 0.13 °F/decade · sea level 0.06 in/yr → 0.11–0.14 in/yr (since 1993) · oceans 70% · Lyme 3.74→7.95 per 100,000 (1991–2014) · West Nile <1% · birds >40 miles north since 1966 · 250,000 extra deaths/yr (2030–2050) · USD 2–4 bn/yr by 2030 · Chicago 1995 (Jul 11–27)
L6	Transport 20–25% world energy · 87% fossil power · coal 39→28%, oil 40→33%, gas 15→24%, renewables 3% · OECD transport share 24.2% (1971) → 33.7% (2016) · oil fuels 94%: gasoline 43.5, diesel 34.5, other 21.8 · transport 22% global GHG · 0.8 °C rise
L7	IARC 1989, CARB 1998 · children 50% higher risk · 1.17 m road deaths, 10 m injured · US 45,000 (2006) · CO/benzene in car 2–5× · diesel truck 50–100× particles · LA 60% more, Clean Air Act 1970, smog 1940s for 3 decades · 1% NMT shift → 2–4% less emissions · 1-cent tax → −0.25% / −0.26% · 20% price → 2,000 + 600 deaths · smart growth 20–40% less driving · speed <20 km/h · hotspots 26–28% time = 54–56% dose · P90 · 1-min PM2.5
L8	1 in 10 amphibian death rate · Olivia 1400 t IFO, 70 t MDO, ~3700 penguins · wheat India −36% (2010 vs 1980s), 50% some states, 90% from short-lived pollutants · underpass fence 1.5 m
L9	>200 m chemical effects · Haven 1991 144,000 t, >10,000 t spilled · Gulsar Ana 2009: 39,250 MT cargo, 568 MT HFO, 66 diesel, 8 lube; 120 days, 7500 man-days · RENA 2011: 1368 containers, 1700 t IFO 380, 61 t diesel, ~200 t released, >1000 + 250 birds · burning to 1 mm
L10	vehicles ≈ 70% traffic noise · CPCB: sensitive 50/40, residential 55/45, commercial 65/55, industrial 75/70 · p_0 = 20 µPa; 1 B = 10 dB · Leq example 94.3 vs 93 dBA · ear 500 Hz–8 kHz · barrier 5 + 3 = 8 dB ≈ ÷6 traffic · heights 1–2.5 / 4–10 / 2–5 m · 10 kg/m² · 30 dB TL = 99.9% · c = 1125 ft/s (343 m/s) · example: l = 27.3 ft, a = b = 17 ft, 6.7 ft, N = 0.61, IL ≈ 10 dB

Quick difference table

Pair	Difference
Weather vs climate	hours vs hundreds–millions of years
Smart Cities vs AMRUT	core infrastructure (100) vs project approach basic services (500 cities)
Frequency vs Regularity vs Adequacy	how often / punctuality / capacity
Booms vs skimmers vs sorbents vs dispersants	contain / recover / absorb / break into droplets
Transmission loss vs insertion loss	through barrier vs with-vs-without barrier
Road drainage vs embankment	lowers vs raises water table
Stakeholders vs customers/suppliers	stakeholders are neither

Part 4 — Practice MCQs (3 per lecture = 30)

Answer key with slide references follows the questions.

Q1 (L1). The Cheshunt Railway (1825) is described as the world's first monorail drawn by:
A. steam engine B. a horse C. electric motor D. wind power

Q2 (L1). The Rover Safety bicycle's chain drive meant that:
A. both wheels could be the same size B. the front wheel had to be larger C. riders paddled their feet on the ground D. pedals were on the front axle

Q3 (L1). Concorde first flew, entered commercial service and operated till respectively:
A. 1959, 1966, 2000 B. 1969, 1976, 2003 C. 1966, 1969, 2003 D. 1976, 1987, 2003

Q4 (L2). According to Kadiyali, how many factors influence the efficiency of a transportation system?
A. 5 B. 8 C. 11 D. 15

Q5 (L2). In the Sussman components, "people/organizations who are not customers or suppliers but are nonetheless concerned" are:
A. Competition B. Supply industry C. Government D. Stakeholders

Q6 (L2). Which is NOT stated as an alternative name for multimodal transport?
A. Combined transport B. Intermodal transport C. Unimodal transport D. Mixed-modal transport

Q7 (L3). The Dedicated Freight Corridor (India) is expected to save more than ___ in its first 30 years:
A. 45 million tons of CO_2 B. 450 million tons of CO_2 C. 4.5 billion tons of CO_2 D. 750 million tons of CO_2

Q8 (L3). The Johannesburg BRT is described as:
A. the first full BRT system in Africa B. the first BRT in the world C. a system costing US$900 million D. a response to the 2020 Olympic Games

Q9 (L3). Hong Kong's 93% of trips are made by public transport (48%) or:
A. cycling (45%) B. walking (45%) C. walking (48%) D. private cars (45%)

Q10 (L4). Which scheme covers 500 cities with population of 100,000 or more and a "project approach"?
A. NUTP B. 100 Smart Cities C. AMRUT D. NURM

Q11 (L4). The slide on infrastructure gap gives registered-vehicle increase and urban road increase over the last decade as:
A. 124% and 219% B. 219% and 124% C. 438% and 124% D. 219% and 438%

Q12 (L4). Which statement about NMT policy in India is stated in the slides?
A. Many projects were aimed exclusively at NMT B. NUTP vision says NMT should be the last-mile connector C. JNNURM was adopted in 2009 D. NMT share rose in all cities since the 1980s

Q13 (L5). Since 1993 the average sea level has risen at:
A. 0.06 in/yr B. 0.11–0.14 in/yr C. 0.13 °F/decade D. 0.8 in/yr

Q14 (L5). Neuroinvasive West Nile cases occur in:
A. over 50% of infected people B. about 10% C. less than 1% D. exactly 5%

Q15 (L5). According to the slides, the warmest year on record is:
A. 2019 B. 2020 C. 2014 D. 2016

Q16 (L6). The transportation sector accounts for what share of global GHG emissions?
A. 12% B. 22% C. 33.7% D. 87%

Q17 (L6). Oil-related fuels account for ___ of energy used by transportation modes, with gasoline the most used at ___:
A. 94%; 43.5% B. 87%; 34.5% C. 94%; 34.5% D. 43.5%; 94%

Q18 (L6). The Environmental Kuznets Curve hypothesis says that:
A. pollution always rises with income B. pollution increases early, then the trend reverses beyond some per-capita income C. pollution falls first then rises D. income has no link with pollution

Q19 (L7). A 1% shift from motorized to non-motorized modes can typically reduce emissions by:
A. 0.25% B. 1% C. 2–4% D. 20–40%

Q20 (L7). CO and benzene levels inside cars can be around ___ than at the roadside:
A. 2–5 times higher B. 2–5 times lower C. 50–100 times higher D. equal

Q21 (L7). Smart growth residents typically drive:
A. 2–4% less B. 20–40% less C. 50% less D. 60% more

Q22 (L8). Which is NOT one of the four main effects of transport infrastructure on wildlife?
A. Habitat loss B. Traffic mortality C. Population subdivision D. Ecological disequilibrium

Q23 (L8). Wheat yield in India in 2010 compared with the 1980s was:
A. 36% higher B. 36% lower C. 90% lower D. 50% lower nationwide

Q24 (L8). "Extensive bridges" as a crossing structure are meant for:
A. frogs and toads B. turtles C. large carnivores like grizzly bears and wolves D. stone martens only

Q25 (L9). The HAVEN tanker (1991, Italy) spill happened because:
A. it ran aground on a reef B. it exploded due to an electric spark during tank cleaning C. it collided with another ship D. ballast water leaked

Q26 (L9). A "tube recharge well" is known in Gujarat, India as:
A. Bhungroo B. Ijzeren Vrouw C. Astrolabe D. Wakashio

Q27 (L9). Regarding acid rain's effects on humans, the slides state:
A. only direct effects B. only indirect effects C. both direct and indirect equally D. no effect

Q28 (L10). Combined effect of a noise barrier (5 dB + 3 dB = 8 dB) is equivalent to reducing the traffic by a factor of:
A. 2 B. 3 C. 6 D. 8

Q29 (L10). L90 represents:
A. sporadic noise B. median noise C. background/ambient noise D. peak noise

Q30 (L10). CPCB (India) night-time limit for sensitive areas such as parks, schools, hospitals is:
A. 50 dB(A) B. 40 dB(A) C. 45 dB(A) D. 55 dB(A)

Answer key

Q	Ans	Source-based explanation
1	B	"Drawn by a horse" — L1 s19
2	A	Chain drive → both wheels same size — L1 s27
3	B	1969 / 1976 / 2003 — L1 s42
4	C	"11 Factors" — L2 s18
5	D	Stakeholders definition — L2 s32
6	C	Only "Combined" and "Intermodal" named; "Unimodal" not stated — L2 s33
7	B	>450 million tons CO_2 in 30 years — L3 s34
8	A	First full BRT in Africa — L3 s36
9	B	Public transport 48% + walking 45% — L3 s31
10	C	AMRUT for 500 cities, project approach — L4 s23
11	B	219% vehicles vs 124% roads — L4 s10
12	B	NUTP vision: NMT last-mile connector — L4 s34 (s35: no direct NMT-only projects; JNNURM was 2005)
13	B	0.11–0.14 in/yr, ~twice the long-term trend — L5 s21
14	C	<1% — L5 s31
15	D	2016 warmest; 2020 2nd — L5 s8
16	B	22% — L6 s14
17	A	94% oil-related; gasoline 43.5% — L6 s13
18	B	EKC — L6 s27
19	C	2–4% — L7 s36
20	A	2–5 times higher — L7 s13
21	B	20–40% less — L7 s41

22	D	Ecological disequilibrium is an indirect impact; the four are habitat loss, traffic mortality, population subdivision, barrier effect — L8 s3, s5
23	B	36% lower — L8 s36
24	C	Extensive bridges for grizzly bears and wolves — L8 s43
25	B	Electric spark during tank cleaning — L9 s18
26	A	Bhungroo in Gujarat — L9 s36
27	B	"Only indirect effects on humans" — L9 s29
28	C	Factor of 6 — L10 s34
29	C	L90 = background/ambient — L10 s30
30	B	CPCB night 40 dB(A) for sensitive areas — L10 s9

Part 5 — Coverage Audit

Method: each PDF's text layer was extracted slide by slide and read. Slides with no usable text (pictures, charts, tables saved as images) were then rendered and read visually; their content is in Part 2A. Values marked ≈ are approximate readings of unlabeled charts. Photo-only slides contribute captions only.

PPT filename	Total slides	Slide numbers covered in notes	Extraction issues
lec 01.pdf	50	1–48 (49 references, 50 thank-you)	s47 read from rendered image (Part 2A), mirrored in PDF so item positions approximate. Slide 25 title (1876) conflicts with body (1879) – kept both.
lec 02.pdf	38	1–36 (37 references, 38 thank-you)	Photo slides: captions only.
lec 03.pdf	45	1–43 (44 references, 45 thank-you)	s16, 19, 22, 26–29 read from images (Part 2A). s24 SDG pairing as on slide. s38 US column inferred from order. s36 "(2026)" kept as written.
lec 04.pdf	44	1–42 (43 references, 44 thank-you)	s5, 6, 8, 9, 17, 18, 24, 31, 32, 36, 38–41 read from images (Part 2A); s6, 31, 38–41 give approximate bar readings (no printed values).
lec 05.pdf	44	1–42 (43 references, 44 thank-you)	s7, 10, 11, 13, 23, 24, 36–40 read from images (Part 2A); s7, 11, 24, 38, 39 approximate.
lec 06.pdf	30	1–28 (29 references, 30 thank-you)	s5, 7, 9, 15, 18, 20, 24, 25 read from images (Part 2A); s15 bar values approximate.
lec 07.pdf	49	1–47 (48 references, 49 thank-you)	s6, 7, 14, 30, 31, 32 read from images (Part 2A). s29 age-limit symbols (≥, <) lost in text layer: ages not guaranteed.
lec 08.pdf	51	1–49 (50 references, 51 thank-you)	s4, 6, 8, 13, 16 read from images (Part 2A); s13 values approximate. Photos s39–48: captions only.
lec 09.pdf	53	1–51 (52 references, 53 thank-you)	s5, 30, 46 read from images (Part 2A). Photos s31–42: captions only. Equation arrows lost in text layer; reproduced as written.
lec 10.pdf	51	1–10, 13, 17–49 (50 references, 51 thank-you)	**NOT recoverable: s11, 12, 14, 15, 16** (traffic-noise factor lists absent from the PDF; headings + pictures only). s9, 10, 19, 20, 23, 26, 44, 46, 48 read from images (Part 2A).

Totals: 10 files, 455 slides (50+38+45+44+44+30+49+51+53+51). All 10 PPTs were processed.

Quality checks performed

- **Check 1 (files):** all 10 PDFs processed.
- **Check 2 (slides):** every slide's extracted text read; image-only slides listed above.
- **Check 3 (no outside info):** a formula for insertion loss that I initially wrote from memory was removed; notes keep only slide wording. Full forms of acronyms are given only where on slides (e.g. AMRUT, FHWA, CPCB, BRT as on slides); other abbreviations left as-is.
- **Check 4–6:** numbers, units and qualifiers re-read against the extracted text while writing; ambiguities marked " [as written]".
- **Check 7 (PDF readability):** done after rendering (see delivery message).
- **Not recoverable:** L10 slides 11, 12, 14, 15, 16 (no factor text in the PDF). Chart values marked ≈ are approximate. Slide L10 s44 and a few photo slides were checked only for labels.